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Do bank CEOs learn from banking crises?☆

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ABSTRACT

Does the early-career exposure of bank CEOs to the 1980s savings and loans (S&L) crisis affect the outcomes of banks they subsequently managed? We measure the S&L crisis exposure by the bank failure rate in the states where CEOs worked during the S&L crisis. Armed with this measure, we find that banks managed by CEOs with higher S&L crisis exposure took on less risk and that these banks better survived the financial crisis of 2008. In particular, CEOs adjusted risk attitudes in areas causing the S&L crisis: their more intense crisis experience reduced banks' interest rate risk, exposure to risky financial innovation and credit risk. We establish the causal interpretation of the findings by evaluating the impact of crisis exposure via CEO hometown states and exploiting quasi-exogenous turnovers due to CEO retirement. Overall, CEOs learned from the past industry crisis which helped curtail their institutions' risk exposures and enhance later crisis performance.

1. Introduction

One commonly held notion is that financial institutions' substantial exposure to risk caused the financial crisis in 2008. Yet some banks fared well during this crisis. Is there something unique about the risk taking seen at these banks? There has been considerable research on the determinants of bank risk-taking behavior and performance. Prior work has focused on aspects such as financing (Brunnermeier, 2009), governance (Fahlenbrach and Stulz, 2011), and culture (Ellul and Yerramilli, 2013; Fahlenbrach et al., 2012). However, despite the considerable academic work devoted to identifying how early-life experiences of executives matter for corporate policies (Malmendier et al., 2011; Dittmar and Duchin, 2016; Bernile et al., 2017; Schoar and Zuo, 2017) and policymakers' emphasis on the key role of leadership in shaping bank behavior (Dudley, 2014), the literature has overlooked the influence of CEO experiences on bank performance and risk-taking behavior. This paper fills this gap by exploring how the early-career experiences of the 1980s savings and loan (S&L) crisis impact banks and whether bank CEOs have learned from their experiences.[1]

Following the experience-based learning literature (Malmendier, 2021), our main hypothesis — the *learning hypothesis* — is that banks whose CEOs have a higher exposure to past banking crises take less risk and have better crisis performance. The reason is that these CEOs actively learn about a crisis scenario from the intense and rare experience and become more cautious about areas that caused the systemic fallout. Notably, such learning and adjustment of risk attitudes by bank leaders are critical for the banking sector: neglecting risk embedded in prevalent industry practices (e.g., eroding lending standards) can lead to financial crises which are essentially the ex-post realization of systemic risk (Shleifer, 2011; Acharya et al., 2017).

Alternatively, CEO crisis experience could have the opposite or no impact on banks. For example, CEOs with higher exposure to the S&L crisis could engage in gambling for resurrection behavior in the spirit

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[1] By banks, we refer to bank holding companies (BHCs).

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of Bernile et al. (2017), which then exacerbates the risk taking of banks. It is also possible that CEOs are hired for implementing a predetermined strategy without imposing their preference on banks, and hence their prior experiences are irrelevant for bank outcomes. Identifying relevant factors of risk taking in connection with CEO experiences can shed light on building a financial system less vulnerable to macroeconomic shocks.

The 1980s S&L crisis is a compelling setting for studying the implications of experience on bank CEO behavior. The S&L crisis led to an unprecedented surge in bank failures. At least 2920 banks and S&Ls from all across the US were closed or received Federal Deposit Insurance Corporation (FDIC) financial assistance. The sheer scale and intensity of the banking crisis affected the real economy, financial markets and the subsequent regulatory environment. It would therefore not be surprising if the S&L crisis left an indelible imprint on future professionals in the banking industry who experienced the crisis first-hand. Moreover, the parallels between the S&L crisis and the recent 2008 financial crisis suggest that bank CEOs may have been able to draw valuable lessons from the S&L crisis that were relevant for the later financial crisis. Finally, the S&L crisis spanned several years and affected nearly all US states. Its state- and time-varying aspects create rich cross-sectional differences in crisis exposure of CEOs for empirical investigation.

To test whether the S&L crisis experiences of a bank CEO matter for the bank that she manages several years later, we relate the bank failure rate in the state where the CEO worked during the S&L crisis to the outcomes of the bank under her charge in the later years. We define *Crisis Exposure* as the logarithm of the highest state-level annual bank failure rate experienced by the CEO during the S&L crisis, where the failure rate is the ratio of failed deposits to the lagged total bank deposits. This state-level measure is less susceptible to endogeneity concerns compared to an alternative employer-level measure. The reason is that the state-level shocks are more likely beyond the control of CEOs, and hence it is less likely that unobserved factors are driving both the CEOs' exposure to the S&L crisis at the state level and the bank policies that she implements in the later years.

Consistent with the *learning hypothesis*, we find that CEOs with greater S&L crisis exposure (hereafter *conservative* CEOs) are more conservative with regard to risk taking. We estimate how CEO S&L crisis experience matters from a sample of 336 publicly listed banks during the period from 1995 through 2009. Panel regressions show that *conservative* CEOs take on less risk accounting for unobservable bank heterogeneity and time-varying economic conditions at the state level. Specifically, an increase in *Crisis Exposure* is associated with lower systemic risk as measured by equity beta and the marginal expected shortfall metric of Acharya et al. (2017), lower tail risk and lower equity volatility. These findings support the view that the S&L crisis experiences could shape CEOs' attitudes toward risk which are then imposed on banks.

One might imagine that *conservative* CEOs would be particularly mindful of specific areas which previously contributed to the S&L crisis. Given that the interest rate risk, risky financial innovation and credit risk contributed to the escalation of the S&L crisis, CEOs who witnessed the resulting industry fallout could learn the implications of such shocks and implement conservative policies in these areas. We find empirical evidence in support of this conjecture. Banks led by *conservative* CEOs are more resilient to interest rate risk: CEO *Crisis Exposure* is negatively related to the covariance between bank returns on assets and interest rate shocks (Drechsler et al., 2021). Moreover, CEOs who experienced the worst of the S&L crisis operate banks that have *lower* exposure to risky financial innovation like derivative trading, and *lower* credit risk as measured by the share of non-performing loans. Furthermore, we show that the effects of crisis exposure on bank risk-taking decisions are stronger for CEOs who held C-level positions in the banking sector during the S&L crisis and thus had a more salient crisis experience. These various pieces of evidence illustrate the domain-specificity feature of experience-based learning as explained in Malmendier (2021).

Next, we address two major concerns related to the causal interpretation of the aforementioned findings. First, conservative banks may select CEOs who have had greater exposure to the S&L crisis. We address this concern in various ways. Following Bertrand and Schoar (2003), we include bank fixed effects and time-varying characteristics to capture factors that affect a bank's choice of CEO. Examining the dynamic effects, we do not find evidence that pre-existing trends (prior to conservative CEOs taking office) in bank risk and policies are driving our results. Furthermore, we exploit the quasi-exogenous timing of retirement-induced CEO turnovers to causally demonstrate the policy changes accompanied by the change of *Crisis Exposure* around exogenous CEO turnovers. Finally, we match the baseline sample with additional bank CEOs who did not experience the S&Ls crisis based on the bank conditions prior to the CEO appointments and our findings continue to hold. These results suggest that our findings are robust to the endogenous CEO-bank matching process.

A second concern is that a CEO could choose her employment location in the 1980s based on state characteristics, resulting in a self-selection issue. To mitigate this concern, we study the impact of bank failure rate in the CEO's state of birth during the S&L crisis on the outcomes of banks under her charge later. Since CEOs cannot select the states that they were born in, they have no control over their hometown states' exposure to the crisis. Based on the hometown "bias" studies (Yonker, 2017; Jiang et al., 2019; Lim and Nguyen, 2020) which suggest that CEOs remain connected to their states of birth via information and social networks, we expect that banks managed by CEOs whose states of birth were more exposed to the S&L crisis tend to eschew risk. Indeed, we find this conjecture to be the case empirically.

A natural question to ask is whether the impact of *Crisis Exposure* manifests during the crisis years. As Acharya et al. (2017) explain, crisis performance essentially captures the ex-post realized systemic risk. Given the aforementioned strong negative association between CEO *Crisis Exposure* and bank systemic risk, we expect bank crisis performance to vary with *Crisis Exposure* as well.

Consistent with the *learning hypothesis*, we find that banks managed by *conservative* CEOs deliver better performance during the 2008 global financial crisis. A one-standard deviation increase in *Crisis Exposure* is associated with 4.2% higher stock market return during the crisis years. To put the effect in perspective, the bank crisis return of a CEO with 7% S&L state bank failure rate (the 75th *Crisis Exposure* percentile) is 3.1% higher than that of a CEO with 3.1% failure rate (25th percentile). For operating characteristics, banks under the charge of *conservative* CEOs have higher returns on assets (ROA). Notably, these banks have a lower share of real estate loans and hoard high-quality liquid assets in the spirit of domain-specific learning. The rationale is that *conservative* CEOs were exposed to the bad outcomes of imprudent lending (especially real estate lending) and the widespread evaporation of liquidity in the 1980s, and would likely be more judicious in these two areas. Following the bank failure definition of Fahlenbrach et al. (2012), we also show that banks managed by *conservative* CEOs are more likely to survive the crisis. Specifically, a one-standard deviation increase in *Crisis Exposure* is associated with 2.3% lower probability of failure between 2007 and 2009. Since the unconditional probability of failure during the same period is 6.6% for the sample banks, this effect translates to an economically significant decline of 34.8%. These results corroborate the view that bank CEOs learned from the previous banking crisis and identified risks in time to start taking corrective and precautionary actions of strengthening the safety of their bank.

To mitigate the concern that banks may choose *conservative* CEOs to cope with the 2007–2008 financial crisis rather than CEOs themselves affecting banks' crisis performance, we focus on CEOs who were hired long before the onset of the 2008 financial crisis and weathered its duration. Since banks in these cases are unlikely to predict the arrival and the timing of the 2008 financial crisis, the assignment of *Crisis Exposure* to banks during the recent crisis is plausibly orthogonal to banks' hiring decisions in the distant past. The enhancing effect of *Crisis Exposure* on banks' crisis performance remains robust.

Overall, CEO learning from the S&Ls crisis helped banks curtail risk and enhance crisis performance, but we also evaluate whether alternative CEO incentives could yield similar results. First, *conservative* CEOs could refrain (for personal reasons such as career concerns) from assuming the optimal level of risk exposure for the banks they manage and such risk avoidance could destroy shareholder value. However, we find that operating performance and stock returns are *not* lower for banks with *conservative* CEOs. Second, *Crisis Exposure* might correlate with compensation packages which provide CEO incentives to prefer a certain level of systemic risk. Yet there exists no statistically meaningful association between *Crisis Exposure* and compensation packages in the subsample of CEOs filtered by data availability. In short, these two alternative incentives are not valid explanations for our findings, which in turn lends support to the *learning hypothesis*.

Finally, we perform a battery of robustness tests. The main findings of the paper hold for both large and small banks measured by asset size and are robust to controlling for other crises. We reproduce the main findings using (1) alternative measure construction of outcome variables; (2) the binary version of *Crisis Exposure*; (3) an alternative period for defining the S&L crisis; (4) alternative model specifications; and (5) an alternative sample period of 1995–2015.

This paper provides empirical support for domain-specific learning based on experiences, whereby individuals hesitate to repeat actions that previously led to painful outcomes. In doing so, this paper contributes to the finance literature on experience-based learning (Malmendier, 2021). Theoretical work in this area demonstrates that experiential learning by investors can have asset pricing implications (Collin-Dufresne et al., 2017; Ehling et al., 2018). Empirical work on this topic shows how past experiences of investors shape their inflation expectations (Malmendier and Nagel, 2016), portfolio choices (Greenwood and Nagel, 2009), trading performance (Seru et al., 2010) and allocations to IPOs (Chiang et al., 2011). However, except for Greenwood and Nagel (2009) who focus on mutual fund managers, these studies typically analyze individual investors. Despite the importance of banks to the financial sector and the key role played by bank CEOs, none of these papers study bank CEOs. The findings fill this critical gap in the literature. Importantly, this paper is among the first to causally demonstrate that professionals' experienced-based learning from a past crisis benefits their future institutions' subsequent crisis performance.

The results also resonate with the literature on bank behavior, particularly during financial crises. Research has shown that bank behavior is driven by bank-specific factors such as liquidity (Cornett et al., 2011), securitization (Loutskina, 2011; Erel et al., 2014), short-term capital market financing (Irani and Meisenzahl, 2017), and risk culture (Ellul and Yerramilli, 2013). Some papers speak to the experiences of banks: Bouwman and Malmendier (2015) find that banks that survived periods of undercapitalization tend to implement higher equity ratios and take less risk and Fahlenbrach et al. (2012) show that banks that underperformed during the 1998 Long-Term Capital Management (LTCM) crisis also performed poorly during the 2008 crisis. But unlike these authors who leverage bank institutional experience, we emphasize the importance of bank leadership experience. Several policymakers have blamed the 2008 financial crisis on the failure of leadership at banks (e.g., Dudley, 2014). However, academic evidence in this topic is scarce.[2] This paper thus complements existing studies in understanding bank behavior from the perspective of leaders.

Finally, the findings relate to the literature that examines how CEO characteristics such as political affiliation (Hutton et al., 2014), military experience (Benmelech and Frydman, 2015) and early work environment (Schoar and Zuo, 2017) are linked to firm outcomes. Bertrand (2009) provides a comprehensive review. These research papers exclude financial services firms (and utilities) from their analyses and therefore are silent on the possibly unique implications of bank CEO experiences. Our paper proposes a new channel that explains the heterogeneity in bank risk management practices—an aspect missing in the CEO literature. In particular, we show that CEOs that have been shocked by prior banking crises more likely foster a culture of financial prudence in managing interest rate risk and credit risk.

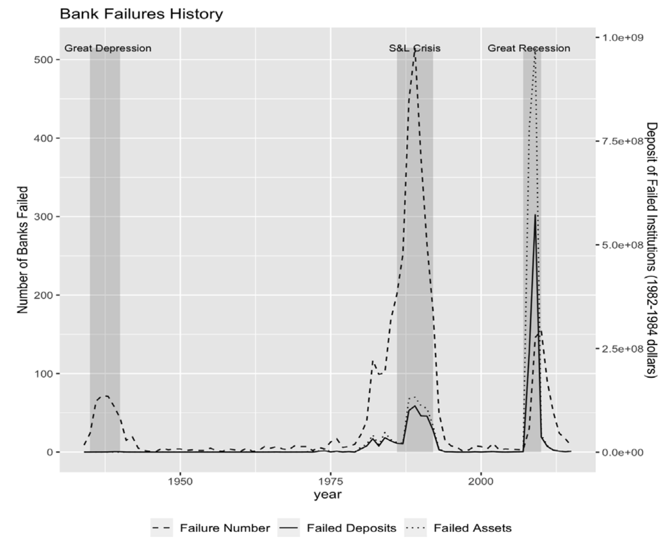

Fig. 1. Number of US bank failures,1930–2015. This figure reports the total number of commercial banks, commercial and savings banks, savings institutions, savings banks and savings associations that were assisted, put under receivership, or closed by the Federal Savings and Loan Insurance Corporation (FSLIC) or the Federal Deposit Insurance Corporation (FDIC) as well as the total amount of failed deposits (assets) from 1930 through 2015. The total amount of failed deposits and assets are inflation-adjusted to 1982-1984 dollars with the annual average of the monthly consumer price index for all items and all urban consumers.
Source: FDIC, Bureau of Labor Statistics and author calculations.

2. Timeline of the S&L crisis

The 1980s S&L crisis witnessed an extraordinary surge in the number of bank failures, with at least 2920 banks and S&Ls either closing or receiving FDIC financial assistance.[3] The unprecedented magnitude of the banking crisis affected both the economy and the financial markets, and in its aftermath regulations tightened in response to the crisis. This event has received extensive media coverage and a large body of academic work has analyzed its causes and consequences.

Fig. 1 reports the total number of commercial banks, commercial and savings banks, savings institutions, savings banks and savings associations that were assisted, placed under receivership, or closed by the Federal Savings and Loan Insurance Corporation (FSLIC) or by the FDIC as well as the total amount of failed deposits (assets) inflation-adjusted to 1982–1984 dollars from 1930 to 2015. It depicts three pronounced peaks in bank failures in the US during that period: the Great Depression of the 1930s, the 1980s S&L crisis and the 2008 financial crisis. Fig. 1 also shows that the number of bank failures during the 1980s S&L crisis dwarfs that of the Great Depression and the 2008 financial crisis, underscoring the significance of the S&L crisis as a landmark event for the banking industry.

[2] Related to studies on bank CEOs, Ho et al. (2016) study the implication of CEO overconfidence on bank behavior.

[3] For the details on the 1980s S&L crisis, please refer to the FDIC summary available at https://www.fdic.gov/bank/historical/history/vol1.html.

2.1. Background

Several factors set the stage for instability in the banking sector during the 1970s. During that period several currencies moved from a fixed-rate regime to a floating-rate regime, and as a result exchange rates became more volatile globally. In response to the oil embargo and other supply shocks, oil prices increased drastically. Concomitantly, interest rates varied wildly in response to inflation and expectations about inflation as well as the anti-inflationary monetary policies adopted by the Federal Reserve.

When the Federal Reserve doubled the federal funds rate to reduce inflation in 1979, it further undermined the financial health of the thrift industry. Congress reacted by deregulating the thrift industry and passing two laws: the Depository Institutions Deregulation and Monetary Control Act of 1980 and the Garn-St Germain Depository Institutions Act of 1982. The deregulation of deposit interest rates exerted upward pressure on banks' funding costs because both banks and thrifts relied on short-term funding; they had to compete for funds by offering higher rates to attract deposits. As the revenue from long-term, fixed-rate mortgages did not vary with short-term interest rates, this squeezed the profits of thrifts and commercial banks alike. Losses began to mount for these institutions.

At the same time, financial innovation reduced the profit margins in traditional banking. Banks faced increased competition from new financial instruments, including money market mutual funds, the commercial paper market and securitization. As a result, many banks shifted funds to commercial real estate lending, which featured higher returns but also greater risks. Some banks participated in leveraged buyouts and off-balance sheet activities. Futures, junk bonds, swaps and other new financial instruments also facilitated greater risk taking by banks. Deposit insurance exacerbated the moral hazard problem faced by banks because insured depositors had little incentive to discourage banks from taking excessive risks.

As a result of such risky behavior, thrifts and commercial banks began to suffer extensive losses. During the 1980s, the performance ratios of banks of all sizes deteriorated substantially while their assumed risks and loan charge-offs rose dramatically. Systemic distress followed, with many S&L customers going bankrupt and defaulting on their loans. The S&Ls that had overextended themselves were forced into insolvency and a wave of bankruptcies ensued. The FSLIC was required to repay depositors. From 1986 to 1995, 1043 of the 3234 S&Ls and more than 1600 banks insured by the FDIC were closed or received financial assistance. The overall cost to taxpayers was estimated to be in excess of US$124 billion. In 1991, Congress passed the Federal Deposit Insurance Corporation Investment Act (FDICIA), which recapitalized the FDIC's Bank Insurance Fund and reformed the deposit insurance and regulatory system.

2.2. Geographic coverage

There is significant heterogeneity in the scale of the crisis across states. Table A1 of the Appendix reports each state's total number of failures, total amount of failed assets and total amount of failed deposits during the period from 1980 to 1993. The incidence of bank failure peaked in different years across different states. Fig. 2 plots the time series of the amount of deposits belonging to the failed institutions in four sample states with the highest total failed deposits during the period from 1980 to 1993. Fig. 2 reveals significant variation in bank failures across states and across time.

The domino effect caused serious strains on the FDIC's deposit insurance fund and transformed what might have been a localized issue into a national problem. Regional and sectoral downturns exacerbated the banking crisis, with bank failures traced back to (i) economic downturns related to the collapse in energy prices (Alaska, Louisiana, Oklahoma, Texas and Wyoming), (ii) real-estate downturns (California, the Northeast and the Southwest), (iii) the agricultural recession of the early 1980s (Iowa, Kansas, Nebraska, Oklahoma and Texas), (iv) an influx of banks chartered in the 1980s (California and Texas) and mutual-to-stock conversions (Massachusetts), (v) bans on branching, limited access to geographically diversified loans and funding of growth via core deposits (Colorado, Illinois, Kansas, Texas and Wyoming), and (vi) the failure of a single large bank (Iowa) or a small number of relatively large banks (New York and Pennsylvania).

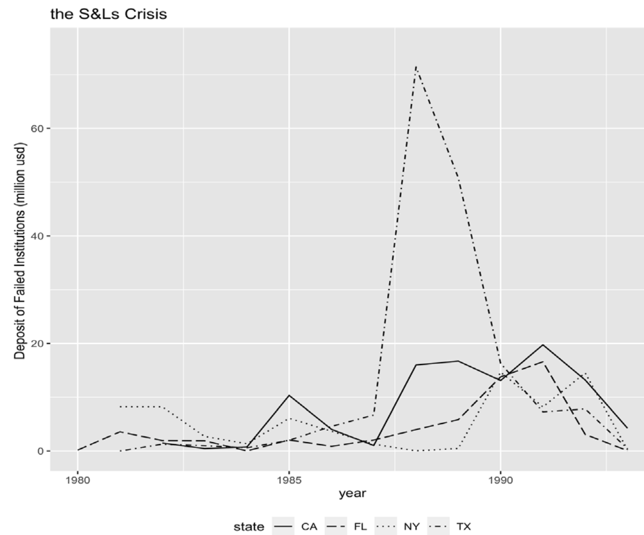

Fig. 2. Example of bank failure time series during the S&L crisis. This figure plots the amount of failed deposits during the period 1980–1993 in California (CA), Florida (FL), New York (NY) and Texas (TX). These four states had the highest total failed deposits.

2.3. Why the S&L crisis?

The S&L crisis is a compelling setting for studying bank CEO crisis experience effects. First, this intense crisis is likely to have left an indelible impression on future bank leaders who experienced it first-hand. Second, the 1980s S&L crisis and the financial crisis of 2008 had similar causes and consequences. They were both preceded by a real estate bubble and a credit boom and led to historically high bank failure rates and unprecedentedly large-scale government bailouts. In addition, both crises featured asset fire sales, liquidity crunches and extreme financial market turbulence. Therefore, bank CEOs may have been able to draw valuable lessons from the S&L crisis that were relevant for the 2008 financial crisis. Third, the state- and time-varying aspects of the S&L crisis create rich cross-sectional differences in the crisis exposure of bank CEOs for empirical tests. Fourth, since considerable time elapsed between the two financial crises, a significant fraction of CEOs would have likely switched firms during that interval, alleviating concerns that the crisis exposure measure captures a persistent firm effect as opposed to a CEO effect.

3. Data and methodology

3.1. Sample construction and data sources

To derive our sample of banks and bank CEOs, we first obtain public bank holding companies (BHCs) that intersect in the FR Y-9C, the Center for Research in Security Prices (CRSP), Standard & Poor's Compustat Bank databases and BoardEx from 1995 through 2009. We exclude years before 1995 to ensure the full recovery of the banking sector from the S&L crisis. The sample stops in 2009 because the financial regulatory landscape underwent considerable changes after the 2007–2008 financial crisis (e.g., the allowed business scope and required core capital for banks). Thus, the period from 1995 through

2009 shall serve as a cleaner and more informative testing period.

Specifically, we begin with 9829 bank holding companies that existed between 1995 and 2009 in FR Y-9C reports. We match these banks to those listed by CRSP based on the PERMCO-RSSD linking table provided by Federal Reserve Bank of New York and our complementary manual matching using bank names.[4] We find 987 matched banks and link all of them to Compustat. We next exclude banks that cannot be matched to BoardEx because we are interested in CEOs' experiences. For this step, we use the linking table of firm identifiers between CRSP-Compustat and BoardEx and select the PERMCO with the highest matching score for each BoardEx firm identifier.[5] We find 481 banks matched to BoardEx and identify 644 bank CEOs. We require banks to have non-missing values of book assets (BHCK2170) and more than one year's worth of observations, which gives us a sample of 452 banks with 605 CEOs.

To identify bank CEOs with exposure to the S&L crisis, we focus on CEOs with available employment histories during the 1985 to 1990 period (see Section 3.2 for the rationale for focusing on these years as representative of the S&L crisis). We find 471 such CEOs. For them, we further identify their employment locations during the 1985 to 1990 period through BoardEx and manual search, which excludes 45 CEOs with missing locations. In total, the final sample comprises 2966 bank-year observations from 336 publicly listed banks and 426 affiliated CEOs.

We gather biographical data on the CEOs from BoardEx, encompassing the names of previous employers, dates of employment, employment tenure, job titles, degrees, alma-mater, gender and date of birth. Marquis Who's Who and Internet search (sources include CEO interviews, alumni associations, obituaries, etc.) provide information on the state of birth for a subset of CEOs. We obtain stock returns from CRSP. The FR Y-9C reports provide BHCs' information on the consolidated balance sheet, income statement, usage of derivatives, breakdown of loan portfolios, security holdings, and regulatory risk capital. To prevent outliers from biasing the results, we winsorize all variables at 1% and 99% in all the analyses.

3.2. CEOs' S&L crisis exposure

Our core explanatory variable *Crisis Exposure* captures CEOs' exposure to the S&L crisis. Following the banking crisis definition of Laeven and Valencia (2008), we take 1988 as the peak of the S&L crisis and choose the 1985 to 1990 period as our S&L crisis period (our results are robust to using 1980 to 1993 as the period for constructing the *Crisis Exposure* measure).

We assume that during the S&L crisis period, the CEO was located in the state in which the CEO's then employer was then headquartered. We first search for CEO employment history from BoardEx and conduct a manual search on the Internet to obtain missing information. For each of the sample CEOs, we have a series of state-year records detailing their employment locations during the 1985 to 1990 time period.

Notably, one-third of the sample CEOs (140 out of 426) moved to a different state after the S&L crisis—i.e., the headquarter state of the bank they manage after the S&L crisis differs from the state they worked in during the S&L crisis. In addition, between 1985 and 1990, 118 CEOs switched states at least once with the number of years a CEO spent in a given state ranging from 1 to 6. Hence, CEOs' intrastate mobility and their non-overlapping time windows associated with the same state during the S&L crisis provide the variation of CEOs' S&L *Crisis Exposure*. Importantly, our main analyses exploit within-bank variations resulting from CEO turnovers. The sample encompasses a total of 92 CEO turnovers from 84 banks. Of these turnovers, 55 from 51 banks involve two CEOs working in different states during the S&L crisis. 20 (17) include two CEOs experiencing the S&L crisis in the same states at different (same) times.

To derive the state-level *Crisis Exposure* measure, we obtain information on bank failures from the FDIC database. Specifically, we divide failed depository institutions' deposits by the total amount of bank deposits in the previous year at the state-year level.[6] Scaling is used to adjust for local economic conditions and banking development. For a given bank CEO, we analyze the maximum of the failure ratios over all of the CEO's identified state-year records between 1985 and 1990. We then take the natural logarithm of the non-zero maximum failure ratios, as they are highly positively skewed (ranging from around 0.17% at the 1st percentile to 70.1% at the 99th percentile). Our main independent variable of interest is therefore

$$\text{Crisis Exposure}_c = ln\left(\max_t\left(\frac{\text{Failed deposits in employment state}_{it}}{\text{Total deposits in employment state}_{i,t-1}}\right)\right) \tag{1}$$

where the subscript c, t and i denote the CEO, the year and the state (respectively).[7]

3.3. Summary statistics

Table 1 reports summary statistics for the key CEO- and bank-specific variables. The variables are defined in Table A2 of the Appendix. Panel A focuses on the full sample spanning the 1995 to 2009 period. Panel B reports the summary statistics of all the variables used in the cross-sectional tests of the financial crisis.

In Panel A, the failure rate values associated with *Crisis Exposure* (*S&L State Failure Rate*) indicate that the average CEO in our data has witnessed a maximum statewide deposit-based bank failure ratio of 8.5% during the S&L crisis. However, there is significant cross-sectional heterogeneity in bank failure rates across CEOs, with the standard deviation of that failure rate measure being 10.6%. This underscores the rich cross-sectional differences in S&L crisis exposure of the CEOs in our sample. We note that the average CEO in our sample is 56 years old. In addition, 35.1% of them hold a post-graduate degree and 2.4% of them are female. In terms of CEOs' employment rankings and industries during the S&L crisis, 47.2% of them were C-suite executives of banks.

On the bank financial information side, the book value of the median bank's total assets is $1.631 billion. The size distribution with respect to the book value of total assets is highly skewed, ranging from $0.702 billion at the 25th percentile to $5.413 billion at the 75th percentile. Given the skewness of the size distribution, we use the natural logarithm of the book assets which is denoted as *Size*. For a median bank, deposits account for 76.5% of the book value of total assets. The tier-1 ratio has a median and mean value of 11.4% and 12.1%, both of which are well above the minimum regulatory capital requirement under the Basel I and II Accords. Revenue from business segments unrelated to interest rates — a proxy for a bank's reliance on off-balance-sheet activities — accounts for 17.6% of the median bank's

[4] The Federal Reserve Bank of New York lists the linking table at https://www.newyorkfed.org/research/banking_research/datasets.html. For manual matching, we take the universe of all banks in CRSP (SIC codes between 6000 and 6300) and name-match banks to RSSD ID identifiers using a string-matching command in STATA. We then rank all potential matches based on their similarity scores and manually check for the correct matches.

[5] Wharton Research Data Services explains the matching algorithm at https://wrds-www.wharton.upenn.edu/documents/1328/Boardex_CRSP_Compustat_Company_Link.pdf.

[6] For each state-year pair, we also calculate two robustness ratios: the number of failed depository institutions to the total number of banks in the previous year and failed assets to the total amount of assets.

[7] When a CEO experienced a 0% state-wide failure rate from 1985 to 1990 (3% of the sample observations), *Crisis Exposure* takes the natural logarithm of 0.09% based on the lowest non-zero failure rates. Results are robust to different choices of the imputed value such as the natural logarithm of 0.1% or 0.05%.

Table 1
Summary statistics.

Panel A: Full sample								
Variable	Count	Mean	S.D.	p10	p25	Median	p75	p90
CEO-Specific variables								
Crisis Exposure	2966	−2.998	1.135	−3.764	−3.479	−2.939	−2.590	−1.633
S&L State Failure Rate in Crisis Exposure	2966	0.085	0.106	0.023	0.031	0.053	0.075	0.195
CEO Age	2966	55.760	7.059	47.000	51.000	56.000	60.000	64.000
Education	2966	0.351	0.477	0.000	0.000	0.000	1.000	1.000
Female	2966	0.024	0.153	0.000	0.000	0.000	0.000	0.000
S&L Bank Executive	2966	0.472	0.499	0.000	0.000	0.000	1.000	1.000
Financial variables								
Assets	2966	20 518	122 475	395	702	1631	5413	20 546
Size	2966	14.635	1.625	12.887	13.461	14.305	15.504	16.838
Book Leverage	2966	0.909	0.024	0.882	0.898	0.911	0.924	0.934
Real Estate Loans	2966	0.473	0.146	0.294	0.377	0.473	0.578	0.657
Liquid Asset1	2963	0.219	0.113	0.092	0.138	0.205	0.282	0.369
Liquid Asset2	2963	0.369	0.174	0.173	0.248	0.347	0.464	0.595
BM	2957	0.799	1.718	0.345	0.441	0.575	0.792	1.158
Deposit	2849	0.751	0.096	0.622	0.691	0.765	0.822	0.863
Tier-1 Capital	2882	0.121	0.038	0.087	0.099	0.114	0.132	0.158
ROA	2876	0.013	0.014	0.004	0.010	0.015	0.019	0.023
Nonint. Income	2966	0.238	0.277	0.071	0.117	0.176	0.264	0.412
Deriv. Trading	2909	0.273	2.612	0.000	0.000	0.000	0.000	0.037
IR Deriv. Trading	2909	0.221	2.317	0.000	0.000	0.000	0.000	0.021
Maturity Gap	2950	0.027	0.188	−0.201	−0.085	0.022	0.141	0.252
Bad Loans	2504	0.008	0.013	0.001	0.002	0.005	0.009	0.018
Bad Loans/Loans	2504	0.012	0.020	0.001	0.004	0.007	0.013	0.027
Risk and returns								
Annual Return	2966	0.089	0.337	−0.304	−0.109	0.076	0.272	0.496
Abnormal Return	2903	0.034	0.386	−0.382	−0.168	0.019	0.251	0.495
Beta	2903	0.650	0.616	−0.003	0.142	0.522	1.060	1.521
Beta Bank	2957	0.329	0.287	−0.001	0.071	0.264	0.602	0.738
MES	2947	−0.017	0.023	−0.042	−0.021	−0.011	−0.004	0.002
MES Bank	2950	−0.017	0.024	−0.045	−0.022	−0.011	−0.003	0.002
Tail Risk	2955	0.048	0.028	0.024	0.031	0.041	0.055	0.084
Equity Volatility	2957	0.385	0.240	0.190	0.243	0.315	0.431	0.681
Idiosyncratic Volatility	2903	0.347	0.221	0.168	0.217	0.287	0.399	0.584
ROA Volatility	2966	0.009	0.001	0.007	0.008	0.008	0.010	0.010
Beta Libor Shock1	2870	0.071	0.134	0.007	0.019	0.043	0.089	0.159
Beta Libor Shock2	2869	0.078	0.097	0.008	0.020	0.047	0.099	0.179
Interest ROA Beta	2713	−0.016	0.259	−0.199	−0.099	−0.039	0.034	0.194
$Return_{1998}$	2418	0.062	0.138	−0.092	−0.034	0.041	0.132	0.244

Panel B: Summary statistics for financial crisis tests								
Variable	Count	Mean	S.D.	p10	p25	Median	p75	p90
CEO-Specific variables								
Crisis Exposure	273	−3.058	1.094	−3.764	−3.479	−3.023	−2.661	−1.658
S&L state failure rate in crisis exposure	273	0.078	0.102	0.023	0.031	0.049	0.070	0.190
CEO Age	273	56.571	6.981	47.000	52.000	57.000	61.000	64.000
Education	273	0.366	0.483	0.000	0.000	0.000	1.000	1.000
Female	273	0.033	0.179	0.000	0.000	0.000	0.000	0.000
Financial variables								
ROA	252	0.005	0.014	−0.011	0.002	0.008	0.013	0.016
Real estate loans	229	0.536	0.141	0.363	0.449	0.547	0.629	0.703
Liquid Asset1	229	0.173	0.098	0.075	0.107	0.153	0.214	0.304
Liquid Asset2	229	0.320	0.167	0.137	0.202	0.295	0.393	0.532
$Size_{2006}$	273	14.671	1.566	13.006	13.557	14.453	15.512	16.655
BM_{2006}	273	0.560	0.186	0.351	0.435	0.525	0.665	0.801
$Book\ Leverage_{2006}$	273	0.906	0.026	0.878	0.896	0.909	0.922	0.933
$Deposit_{2006}$	273	0.749	0.092	0.651	0.701	0.762	0.816	0.847
$Tier\text{-}1\ Capital_{2006}$	273	0.120	0.037	0.089	0.098	0.112	0.131	0.162
$Bad\ Loans_{2006}$	273	0.004	0.005	0.001	0.002	0.003	0.005	0.009
Risk and returns								
FC Return	252	−0.395	0.301	−0.809	−0.635	−0.403	−0.138	−0.031
FC Abnormal Return	252	−0.225	0.446	−0.846	−0.554	−0.195	0.161	0.323
$Beta_{2006}$	273	0.777	0.688	−0.031	0.123	0.745	1.321	1.684
$Return_{2006}$	273	0.121	0.165	−0.055	0.014	0.097	0.211	0.329
$Return_{1998}$	197	0.052	0.138	−0.097	−0.041	0.028	0.120	0.211

This table reports summary statistics of bank- and CEO-related variables, all of which are defined in Appendix Table A2. Panel A focuses on the full sample spanning the 1995 to 2009 period. Panel B reports the summary statistics of all the variables used in the cross-sectional tests of the financial crisis.

Table 2
CEO S&L crisis exposure and bank risk taking.

	Beta		*MES*		*Tail Risk*		*Equity Volatility*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Crisis Exposure	−0.052***	−0.051**	0.003**	0.003**	−0.002**	−0.002**	−0.022**	−0.021**
	(−2.62)	(−2.28)	(2.30)	(2.31)	(−2.23)	(−2.18)	(−2.21)	(−2.04)
$Size_{t-1}$	0.151***	0.193***	−0.004**	−0.006***	0.000	0.002	−0.007	0.006
	(3.52)	(4.17)	(−2.26)	(−2.70)	(0.13)	(0.92)	(−0.32)	(0.20)
$Return_{t-1}$	0.107***	0.108***	−0.003*	−0.004***	−0.001	−0.001	−0.032	−0.025
	(3.38)	(3.35)	(−1.93)	(−2.60)	(−0.90)	(−0.39)	(−1.58)	(−1.07)
BM_{t-1}	0.033	0.032	−0.000	0.001	0.015***	0.013***	0.149***	0.131***
	(1.28)	(1.20)	(−0.24)	(0.63)	(7.53)	(5.75)	(6.20)	(5.27)
$Beta_{t-1}$	0.480***	0.450***	−0.010***	−0.010***	0.006***	0.006***	0.060***	0.057***
	(20.27)	(18.89)	(−9.62)	(−8.61)	(5.61)	(5.68)	(6.07)	(5.79)
Book Leverage$_{t-1}$	−1.384**	−1.199	0.077***	0.041	0.128***	0.133***	1.292***	1.435***
	(−2.39)	(−1.61)	(2.94)	(1.12)	(4.04)	(3.66)	(4.19)	(3.74)
$Deposit_{t-1}$	−0.190	−0.195	0.009	0.011	0.024**	0.026**	0.227***	0.239**
	(−0.96)	(−0.90)	(0.98)	(1.02)	(2.53)	(2.45)	(2.69)	(2.18)
Nonint. Income$_{t-1}$	−0.200***		0.007**		−0.003		−0.060**	
	(−3.27)		(2.53)		(−0.90)		(−2.28)	
Tier-1 Capital$_{t-1}$		−0.091		−0.048		−0.004		0.059
		(−0.13)		(−1.55)		(−0.13)		(0.19)
Bad Loans$_{t-1}$		0.876		−0.382***		0.510***		4.436***
		(0.52)		(−4.02)		(4.72)		(3.15)
Education	0.146***	0.149***	−0.008***	−0.008***	0.009***	0.010***	0.073***	0.080***
	(3.15)	(3.10)	(−2.90)	(−3.08)	(3.76)	(4.36)	(2.82)	(3.20)
CEO Age	0.004*	0.005**	−0.000	−0.000	0.000*	0.000**	0.002*	0.003**
	(1.81)	(2.41)	(−1.10)	(−1.51)	(1.68)	(2.39)	(1.71)	(2.34)
Female	0.002	0.047	−0.002	−0.004	0.002	0.001	0.054	0.056
	(0.01)	(0.43)	(−0.32)	(−0.71)	(0.57)	(0.37)	(1.35)	(1.18)
Constant	−0.801	−1.674	−0.014	0.061	−0.124**	−0.167***	−1.184**	−1.590**
	(−0.85)	(−1.42)	(−0.33)	(1.04)	(−2.35)	(−2.74)	(−2.32)	(−2.35)
State×Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2531	2104	2531	2104	2531	2104	2531	2104
R^2	0.864	0.871	0.808	0.816	0.867	0.888	0.857	0.873

This table reports the results of panel regressions that investigate the relationship between bank CEOs' S&L *Crisis Exposure* and banks' risk. We estimate the regression specification
$Y_{icst} = \alpha + \beta_1 Crisis\ Exposure_c + \lambda_1 X_{it-1} + \lambda_2 C_{ct} + f_i + f_{st} + \epsilon_{icst}$.
The panel data include one observation for each bank-year pair and span the 1995 to 2009 period. In columns (1) and (2), Y is *Beta*, which is bank's equity beta from a market model of daily returns in excess of three-month T-bills where the market is represented by the value-weighted CRSP index. In columns (3) and (4), Y is *MES*, which is the marginal expected shortfall as defined in Acharya et al. 2017 and measured as the bank's average stock return on the 5% worst performance days for the value-weighted CRSP market index. In columns (5) and (6), Y is *Tail Risk* which is the negative of the average daily returns on the 5% worst performance days for the bank over the year. In columns (7) and (8), Y is *Equity Volatility*, which is the annualized standard deviation of daily stock returns. CEO controls include age, gender (Female) and education. The bank control variables are lagged by one year; they include size (i.e., the natural log of the book value of assets), the annual stock return (*Return*), book-to-market ratio (*BM*), market beta, book leverage ratio, deposit ratio, the ratio of non-interest income to the sum of non-interest income and net interest income (*Nonint. Income*), tier-1 capital ratio and the fraction of non-performing loans (*Bad Loans*). All variables are defined in Appendix A2. All specifications include state-year and bank fixed effects. Standard errors are robust to heteroskedasticity and are clustered at the bank level. Numbers in parentheses are *t*-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

total revenue. For loan default measures, the ratio of non-performing loans to total loans (assets) has an average of 1.2% (0.8%).

The stock return and risk measures indicate that heterogeneity prevails among banks. The median and mean annual stock returns for the sample banks are 7.6% and 8.9% from 1995 to 2009, but there is wide variation: a bank at the 25th percentile has an annual return of −10.9% whereas a bank at the 75th percentile has an annual return of 27.2%. The median and average equity betas of sample banks are 0.522 and 0.650. Banks' equity betas are remarkably varied, ranging from −0.003 at the 10th percentile to 1.521 at the 90th percentile. Marginal Expected Shortfall (*MES*), the left-tail exposure following Acharya et al. (2017), has a mean value of −0.021; thus, for an average bank, the average daily stock market return is −2.1% during the worst 5% of days—in terms of returns—for the value-weighted CRSP market portfolio. The 4.8% mean value of *Tail Risk* suggests that an average bank's average daily stock return during its 5% worst days is −4.8%. The mean and standard deviation of banks' *Equity Volatility* are 0.385 and 0.243.

In Panel B, ROA of banks is 0.5% on average during the 2007–2008 financial crisis. The average ratio of real estate loans to book assets, generally the largest part of bank lending, is 53.6%. An average bank holds 17.3% of its assets in liquid assets. Banks' buy-and-hold returns during the 2008 crisis are −39.5% on average and vary from −80.9% at the bottom decile to −3.1% at the top decile.

The summary statistics for these variables are consistent with those in other studies of US banks (Fahlenbrach et al., 2012; Ellul and Yerramilli, 2013; Ho et al., 2016).

4. Empirical analysis

Section 4.1 connects CEO banking crisis experiences with bank risk taking from 1995 through 2009. Section 4.2 examines the relationship between CEO banking crisis experiences and bank performance in the subsequent 2008 crisis. Section 4.3 presents the heterogeneous effects.

4.1. How does CEOs' banking crisis experience matter?

4.1.1. Do banks managed by conservative CEOs take less risk?

We now test the hypothesis discussed in the introduction. The *learning hypothesis* predicts that CEOs with more intense crisis experiences adjust their risk attitudes and implement conservative policies. To evaluate the conjecture, we first examine the relationship between

CEOs' S&L crisis exposure and bank risk taking during the full sample period while controlling for their particular business activities. Econometrically, panel regressions that span a longer period allow us to include bank fixed effects and control for the bank heterogeneity. For this purpose we use panel regressions of the form

$$Y_{icst} = \alpha + \beta_1 Crisis\ Exposure_c + f_i + f_{st} + \lambda_1 X_{it-1} + \lambda_2 C_{ct} + \epsilon_{icst}. \quad (2)$$

The panel data include one observation for each bank-year pair. In Eq. (2), subscripts i, c, s and t denote the bank, the CEO, the state and the year (respectively). The dependent variables are a series of proxies for bank risk taking and the main independent variable is *Crisis Exposure*. We include state×year fixed effects to control for the local time trend, regulatory changes and demand shocks that can affect bank risk levels, and incorporate bank fixed effects to control for unobservable time-invariant heterogeneity across banks. We further address the issue of omitted variable bias via the inclusion of time-varying bank activity controls that are intended to capture the risk faced by banks and CEO personal traits that the literature has posited may affect their risk preferences. The standard errors are robust to heteroskedasticity and are clustered at the bank level.[8]

Table 2 reports results from the aforementioned regression. We use four broad risk-taking measures that capture banks' risk-related policies. Following Acharya et al. (2017), we measure systemic risk with market *Beta* (standard measure of covariance) and *MES* (the bank's average daily stock returns during the 5% worst days for the market).[9] We follow Ellul and Yerramilli (2013) and construct *Tail Risk* as the negative of a bank's average daily stock returns during its 5% worst days. To gauge the aggregate bank risk, we construct *Equity Volatility* as the annualized standard deviation of daily stock returns over the year.

Across these measure, we consistently find that banks led by *conservative* CEOs take on less risk. In columns (1) and (2), the dependent variable is the CAPM-implied market *Beta*. Based on the coefficient in column (1), a one-standard deviation higher *Crisis Exposure* of the bank CEO is associated with a 5.9% (=0.052×1.135) lower *Beta*.[10] Relative to the sample mean, this represents a decrease of 9.08%. To put the effect in perspective, bank *Beta* of a CEO with 7% S&L state bank failure rate (the 75th *Crisis Exposure* percentile) is 0.046 lower than that of a CEO with 3.1% failure rate (25th percentile). The interpretation is that when the overall equity market performs poorly, on average banks that are managed by *conservative* CEOs suffer relatively less. Conversely, these banks do not perform as well as their industry peers when the market as a whole is in an "up" market. In columns (3) and (4), the dependent variable is *MES*, which measures an individual bank's contribution to the losses incurred during an extreme event by the overall market.[11] The measure is signed such that a higher *MES* value corresponds to a lower systemic risk. In column (3), a one–standard deviation increase in *Crisis Exposure* is associated with a 34 (=0.003 × 10000 × 1.135) basis point increase in *MES*, which corresponds to 20% relative to the sample mean. Columns (5) and (6) report results on *Tail Risk*. Column (5) suggests that a one–standard deviation increase in *Crisis Exposure* reduces *Tail Risk* by 23 basis points. In columns (7) and (8), we test for effects on *Equity Volatility*. The estimate in column (7) shows that *Equity Volatility* goes down by 2.5% following a one–standard deviation increase in *Crisis Exposure*.

We control for bank financial characteristics that the literature has found to affect bank risk. In columns (1)–(4), the significant coefficient for *Size* means that larger banks undertake more systemic risk, which is consistent with Acharya et al. (2017). We also find that banks with higher stock returns in the previous year tend to exhibit high systemic risk. In columns (5)–(8), higher book-to-market and leverage ratios increase banks' tail risk and equity volatility. All columns control for the lagged *Beta* and its statistically significant coefficient estimates suggest that the risk-taking at the bank level can be persistent. Note that in the even columns the point estimate of the coefficient for *Crisis Exposure* remains both qualitatively and quantitatively similar when we control for a different set of controls including lagged *Tier-1 Capital* and *Bad Loans* ratios.

To distinguish the effect of crisis experience, we also control for other CEO traits. First, we control for CEOs' educational attainment in case it makes a difference on CEOs' ability to manage banks (King et al., 2016). We construct a dummy variable *Education* indicating whether the CEO earned a post-graduate degree (master's, MBA or PhD).[12] Second, we control for CEO age because an older CEO might have been exposed to different personal, firm, industry or market environments other than banking crisis experience. This CEOs age control also helps rule out the alternative explanation that risk aversion shifts with age or that management ability accrues over the life cycle.[13] Third, we add a control for CEO gender because prior studies have shown that men tend to have different risk attitudes than women (Barber and Odean, 2001 and Eckel and Grossman, 2008). Empirically, education is positively correlated with risk taking. One explanation is that the greater financial literacy imbued by higher education may lead CEOs to be more confident about their ability to manage risk. The coefficient on *CEO Age* is positive and that on *Female* is insignificant.

These findings are not driven by the unusual period of the 2007–2008 Financial Crisis. In Table A4 in the Appendix, we exclude the years 2007 and 2008 and repeat the analysis of Table 2. The point estimate of *Crisis Exposure* remains statistically significant and has a similar magnitude.

In Table A3 in the Appendix, we consider other risk taking proxies. To the extent that CEOs become particularly wary of industry risks and peer banks' common practices after experiencing the industry-specific S&L shock, we replace the market portfolio with the banking sector in constructing *Beta Bank* and *MES Bank* in order to test whether banks led by *conservative* CEOs differ from industry peers. In columns (3) and (4), we construct *Idiosyncratic Volatility* to measure banks' idiosyncratic risk and *ROA Volatility* to capture the fundamental-based bank risk. Consistent with previous findings, higher *Crisis Exposure* correlates with lower *Beta Bank*, higher *MES Bank*, lower *Idiosyncratic Volatility* and lower *ROA Volatility*.

Overall, *Crisis Exposure* captures a significant effect on bank risk taking, beyond bank- and market-level determinants and other CEO traits. This evidence further confirms the view that the S&L crisis experience shapes CEOs' preference and assessment of risk.

4.1.2. Do CEOs learn to manage risk differently?

We now attempt to identify the specific lessons bank CEOs learned from the S&L crisis and applied to bank policies. Malmendier (2021) point out that past experiences affect risk taking in specific domains in which individuals have had exposure to past realizations (domain specificity). In our setting, *conservative* CEOs were exposed to the bad outcomes of interest rate risk, risky financial innovation and imprudent lending during the S&L crisis. Therefore, we expect *conservative* CEOs to adopt conservative risk management policies in these specific areas.

[8] Results are robust to double-clustering of standard errors by bank and year or bank and state-year. Additionally, we also re-run baseline regressions using block bootstrapping by bank and year and results remain overall similar.

[9] Although market beta also measures systematic risks, we follow Acharya et al. (2017) in using it as a systemic risk proxy.

[10] As reported in Table 1 the standard deviation of *Crisis Exposure* is 1.135 from 1995 through 2009.

[11] Value-at-Risk (VaR) is another widely used risk measure employed by financial institutions. One distinction between Expected Shortfall and VaR is that the former captures all losses—that is, including those beyond the VaR threshold.

[12] In unreported robustness checks, we also control for whether a CEO's alma mater belongs to the Ivy League.

[13] We also control for Depression baby CEOs born during the period 1920—1929, as in Malmendier et al. (2011). Because *CEO Age* and the Depression baby dummy variable are strongly correlated, the specifications control only for the former.

Table 3
CEO S&L Crisis exposure, interest rate risk and credit risk.

	(1) *Beta Libor Shock1*	(2) *Interest ROA Beta*	(3) *Deriv. Trading*	(4) *Nonint. Income*	(5) *Bad Loans*	(6) *Bad Loans/Loans*
Crisis Exposure	−0.008*	−0.089**	−0.007*	−0.010**	−0.002**	−0.003*
	(−1.79)	(−2.35)	(−1.74)	(−1.97)	(−2.00)	(−1.89)
Bank Controls	Yes	Yes	Yes	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2511	2350	2522	2531	2224	2224
R^2	0.505	0.624	0.913	0.915	0.786	0.786

This table reports the results of panel regressions that investigate how CEOs' S&L Crisis Exposure affects banks' resilience against interest rate fluctuations, exposure to financial innovation and credit risk. We estimate the regression specification
$Y_{icst} = \alpha + \beta_1 Crisis\ \ Exposure_c + \lambda_1 X_{it-1} + \lambda_2 C_{ct} + f_i + f_{st} + \epsilon_{icst}$.
The panel data include one observation for each bank-year pair and spans the 1995 to 2009 period. The dependent variable Y in column (1) is *Beta Libor Shock1*, which refers to the bank's stock returns sensitivity to the first difference of a post-AR(2) model fitting of the Libor rates. In column (2), Y is *Interest ROA Beta* which refer to the sensitivity of ROA to federal funds rate changes as defined in Drechsler et al. 2021. Y in columns (3) is *Deriv. Trading*, which is the ratio of bank's total gross notional amount of derivative contracts held for trading to assets. In column (4), Y is *Nonint. Income*, the ratio of non-interest income to the sum of non-interest income and net interest income. In column (5)((6)), Y is *Bad Loans* (*Bad Loans/Loans*), which is ratio of the sum of loans post due 90 days or more and nonaccrual loans divided by assets (loans). CEO controls include education, age and gender (Female). The bank control variables are lagged by one year; they include size (i.e., the natural log of the book value of assets), the annual stock return (*Return*), book-to-market ratio (*BM*), market beta, book leverage ratio, deposit ratio and the ratio of non-interest income to the sum of non-interest income and net interest income (*Nonint. Income*). All variables are defined in Appendix A2. All specifications include state-year and bank fixed effects. Standard errors are robust to heteroskedasticity and are clustered at the bank level. Numbers in parentheses are t-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

Interest rate risk management. We first investigate the resilience of bank business models to interest rate variation in the market. Interest rates are central to the business model of banks, which are traditionally viewed as "maturity transformers". However, it remains inconclusive whether rising interest rates benefit or destroy a bank's value. As described by English et al. (2018), the literature has reported contradictory empirical findings about the sign of the correlation between interest rate shocks and equity value. On the one hand, conventional wisdom holds that the maturity transformers should benefit from interest rate reductions that result in a steeper yield curve because banks are in the business of borrowing "short" and lending "long". Thus, the sign and magnitude of the effect should depend on the resulting maturity gap. On the other hand, the asset value of a bank should decline as long-term interest rates rise because the latter trend offsets the associated net interest income. Recently, Drechsler et al. (2021) show that the deposit franchise allows banks to engage in maturity transformation without taking on interest rate risk. In sum, the net effect of rising interest rates on bank equity value can be positive, negative or zero.

Regardless of the direction of the effect, ex-ante unpredictable interest rate movements can pose destructive risks to banks and generate uncertainty in bank value. This uncertainty contributed to the escalation of the S&L crisis in the 1980s: when the deposit rate ceilings was removed, the ensuing jump in deposit rates pushed up banks' funding costs and thus exposed S&Ls to insolvency. The reason is that many S&Ls had exceptionally large duration mismatches which were previously sustainable in the 1970s with the ceiling in place. This highlighted the importance of managing interest rate risk. We hypothesize that CEOs who witnessed more severe S&L losses are more averse to such uncertainty and will therefore engineer their business models to be less sensitive to changes in the interest rate. Yet a bank's business model is an intangible, overarching concept related to income sources, hedging policies, business segments and so forth. Hence we are unable to specify a metric for such models. We therefore opt for two different approaches.

The first approach uses the bank's market valuation as a proxy for its business model. We choose the 3-month London Interbank Offered Rate (*Libor*).[14] To extract interest rates shocks, we take the *first difference* (subscript "Shock1") of the monthly interest rate time series, which eliminates the time trend and seasonality and therefore stabilizes the mean of that time series. To gauge each bank's exposure to interest rate shocks, we regress the monthly returns on its stock against monthly interest rate shocks by year and obtain interest rate beta *Beta Libor Shock1*.[15]

The second approach estimates the impact of interest rates on banks' cash flow. Building on Drechsler et al. (2021), we estimate each bank's yearly sensitivity of ROA to interest rate changes. To ensure the estimation precision, we require each bank-CEO pair to have at least 16 quarterly ROA observations in total and run recursive regressions. We regress the change in a bank's annualized ROA on contemporaneous and three lagged fed funds rate quarterly changes. We then sum up the four coefficients as *Interest ROA Beta* to capture the cumulative effect over a full year and keep the last quarter's observation. For estimation windows with fewer than 16 observations, we use the full history of the bank-CEO pair in question for estimation.[16]

Finally, we estimate panel regressions in the form of Eq. (2) by regressing the obtained betas on *Crisis Exposure*. In this way we verify whether the intensity of crisis experience is associated with bank resilience to interest rate movements. The results using *Beta Libor Shock1* and *Interest ROA Beta* are reported in columns (1)–(2) of Table 3, where we include both state-year and bank fixed effects. We observe that greater *Crisis Exposure* correlates with lower *Beta Libor Shock1* and lower *Interest ROA Beta*.[17] The effects are economically large: a one–standard deviation increase in *Crisis Exposure* reduces *Beta Libor Shock1* and *Interest ROA Beta* by 0.9% and 10.1% (respectively). These effects are not due to bank heterogeneity or local time trend, and they are robust to excluding crisis years (see Table A4 in the Appendix).

For robustness checks, we construct *Beta Libor Shock2* where interest rate shocks are proxied by the residuals (subscript "Shock2") of the

[14] Results are robust to using the T-bill Eurodollar (TED) spread, Treasury bill yields, credit spreads, etc.

[15] Given the ambiguous direction of interest rate shocks' net effects on bank equity returns, we focus on the coefficients magnitude for these two measures.

[16] We also follow Drechsler et al. (2021) and estimate time-invariant *Interest ROA Beta* using each bank-CEO pair's full history (see Section V.B in Drechsler et al., 2021) and later run cross-sectional tests between these betas and *Crisis Exposure*.

[17] In unreported tables, we find the same results from cross-sectional regressions where time-invariant *Interest ROA Beta* is explained by *Crisis Exposure* and bank controls that are collapsed to the bank CEO level.

Table 4
Endogeneity tests.

	Panel A: Quasi-exogenous CEO Turnovers					
	(1) *Beta*	(2) *MES*	(3) *Tail Risk*	(4) *Equity Volatility*	(5) *Interest ROA Beta*	(6) *Bad Loans/Loans*
Crisis Exposure	−0.103*	0.011***	−0.008**	−0.078*	−0.120**	−0.010*
	(−1.95)	(5.35)	(−2.45)	(−1.96)	(−2.13)	(−1.81)
Bank Controls	Yes	Yes	Yes	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	356	356	356	356	306	300
R^2	0.883	0.895	0.908	0.907	0.859	0.841
	Panel B: CEOs' Home State Bank Failures During the S&L Crisis					
	(1) *Beta*	(2) *MES*	(3) *Tail Risk*	(4) *Equity Volatility*	(5) *Interest ROA Beta*	(6) *Bad Loans/Loans*
Crisis Exposure$_{birth}$	−0.166**	0.003**	−0.005**	−0.033**	−0.022	−0.002*
	(−2.48)	(2.18)	(−2.59)	(−2.11)	(−1.02)	(−1.71)
Bank Controls	Yes	Yes	Yes	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	578	578	578	578	548	480
R^2	0.724	0.859	0.864	0.886	0.655	0.750

This table tabulates the panel regressions results from the endogeneity tests for Table 2. The dependent variables in columns (1)–(6) across the three panels are *Beta, MES, Tail Risk, Equity Volatility, Interest ROA Beta* and *Bad Loans/Loans* (respectively). Panel A exploits the subsample of banks with at least one quasi-exogenous CEO turnover where a CEO retires at the age of 63 or above. In Panel B, the independent variable of interest *Crisis Exposure$_{Birth}$*, analogous to *Crisis Exposure*, identifies CEO hometown states' bank failure rates during the S&L crisis. All regressions control for the same bank and CEO characteristics as in column (1) of Table 2, which are not reported to conserve space. All variables are defined in Appendix A2. The regressions in Panel A include state-year and bank fixed effects; those in Panel B include state-year fixed effects. Standard errors are robust to heteroskedasticity and are clustered at the level of bank in Panel A. Numbers in parentheses are t-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

monthly time series of Libor after fitting with AR(2) models to preclude the possibility of autocorrelation. We also examine banks' balance sheet. Banks are insulated from interest rate changes if their interest-earning assets are perfectly matched with interest-bearing liabilities. Prior literature (e.g., Flannery and James, 1984; Gomez et al., 2021) thus argue that the short-term maturity gap ("income gap") captures interest rate risk. We accordingly construct a 1-year maturity gap as the difference between the assets and liabilities that reprice or mature within one year divided by total assets. This concept essentially measures a bank's net floating-rate assets and the extent to which a bank's net interest income are sensitive to interest rates changes—echoing the cash flow approach.[18] Appendix Table A3 reports consistent findings with *Beta Libor Shock2* and 1-year *Maturity Gap* as dependent variables.

Consistent with the domain-specific learning based on experiences, *conservative* CEOs learned from the S&L crisis in terms of adopting business models that are less affected by interest rate fluctuations which previously contributed to the S&L crisis.

Exposure to financial innovation. We now turn to banks' exposure to financial innovation and expect it to have a negative relationship with *Crisis Exposure*. The reasons are two-fold. First, financial innovation such as derivative trading was prevalent in the banking sector before the onset of the financial crisis. Following the evidence from Section 4.1.1 that *conservative* CEOs take on less systemic risk and manage their banks differently than their industry peers, these CEOs are likely to be judicious about the industry trend involving much risk. Second, *conservative* CEOs are likely to be particularly cautious about financial innovations given their experiences in the 1980s. As mentioned in Section 2.1, financial innovations including alternative mortgage instruments and off-balance sheet activities were partially accountable for the S&L crisis.

Our first proxy for the exposure to financial innovation is *Deriv. Trading* which refers to the ratio of the total gross notional amount of derivative contracts held for trading to the book value of assets. A second proxy is *Nonint. Income*, which captures the ratio of non-interest income (e.g., derivative trading income, underwriting fees, etc.) to the sum of non-interest income and net interest income. Brunnermeier et al. (2020) explain that non-interest revenue is generated by business activities different from the traditional deposit taking and lending functions of banks and is positively associated with systemic risk.

We then estimate the panel regression in the form of Eq. (2) by regressing these two proxies (respectively) on *Crisis Exposure*. Columns (3) and (4) of Table 3 tabulate the results. As expected, the coefficient estimates on *Crisis Exposure* across these columns are significantly negative accounting for the time trend, bank heterogeneity and business activities. Thus, these banks trade fewer derivatives and rely less on non-interest business segments. These findings hold for non-crisis years (see Table A4 in the Appendix). Furthermore, column (7) of Table A3 in the Appendix corroborates these findings using a third measure of *IR Deriv. Trading* (the ratio of the gross notional amount of interest rate derivative contracts held for trading to book value of assets) and shows that banks led by *conservative* CEOs trade less interest rate derivatives.

Consistent with the domain-specific learning based on experiences, *conservative* CEOs learned from the S&L crisis to reduce their banks' exposure to risky financial innovation.

Credit risk management. In columns (5) through (6), we check whether *conservative* CEOs exhibit distinctive ways of managing credit risk. As stated in Section 2, the 1980s witnessed intense competitions among financial institutions. Those that were locked in by rate ceilings assumed greater credit risk in order to boost their profitability and these risk-shifting practices were a proximate cause of S&L failures. We expect CEOs who experienced this part of history to be especially careful

[18] The asset part in maturity gap is directly recorded by BHCK3197 which refers to earning assets that reprice or mature within one year. The liability leg considers interest-bearing deposits that mature or reprice within one year, long-term debt that reprices or matures equal to less than one year, variable rate preferred stock, other borrowed money with maturity equal or less than 1-year, commercial paper, federal funds and repo liabilities.

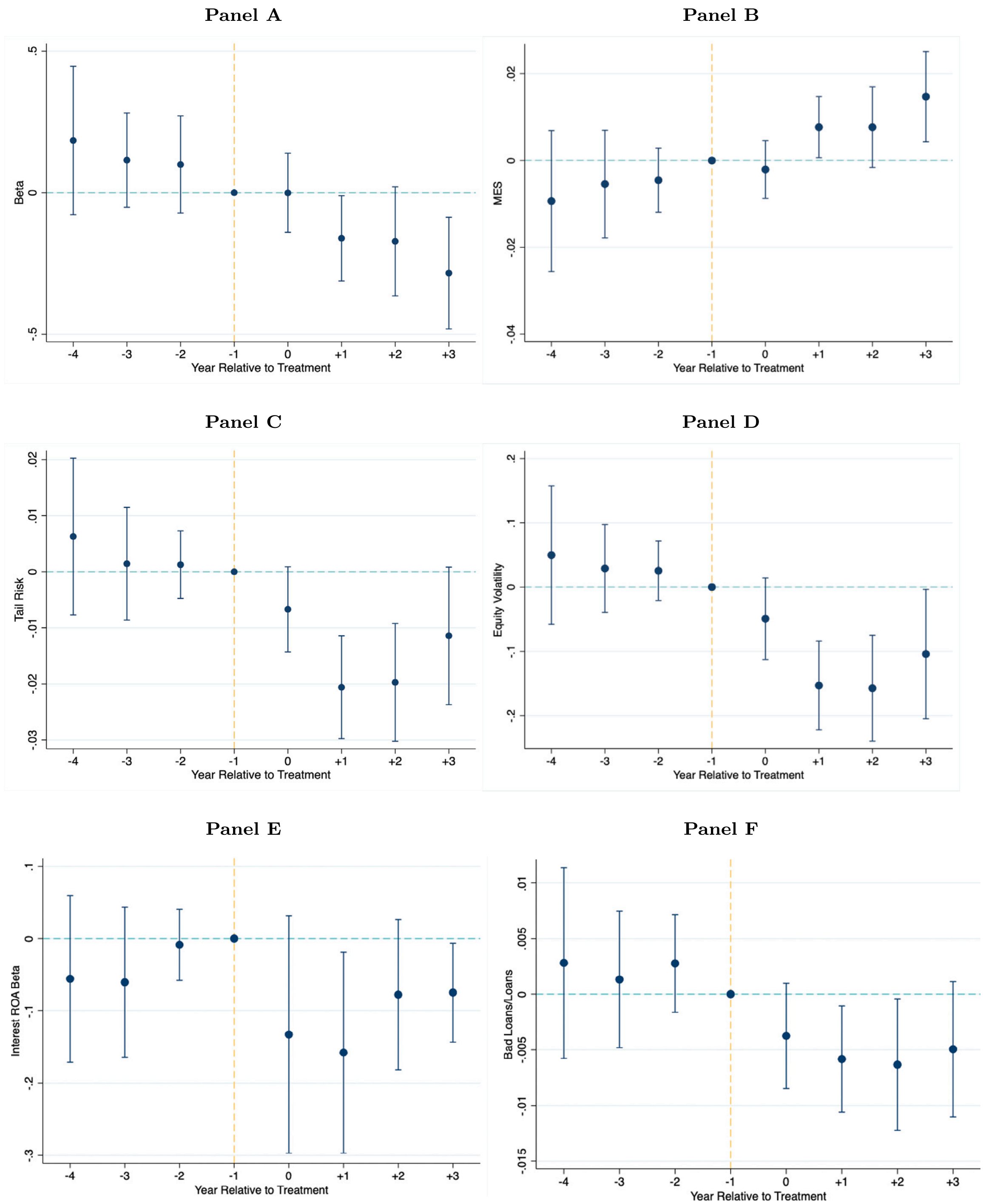

Fig. 3. Timing of change in bank polices. This figure presents event studies in which bank risk measures are regressed on a set of year dummies around the event that a CEO with higher-than-predecessor *Crisis Exposure* takes office. The horizontal axis represents time in years relative to the CEO turnover, while the vertical axis represents the coefficient estimates for event year dummies. The year preceding the turnover is omitted. Each point corresponds to the difference in outcome variables between the given year and the mean during the year preceding the turnover. The outcome variables in Panel A, B, C, D, E and F are *Beta*, *MES*, *Tail Risk*, *Equity Volatility*, *Interest ROA Beta* and *Bad Loans/Loans* (respectively). The specifications include the bank- and CEO-level controls, bank fixed effects, and year fixed effect. Vertical bars represent ninety percent confidence intervals adjusted for clustering at the bank level.

Table 5
CEO S&L crisis exposure and bank performance during the financial crisis.

	FC Return				*ROA*	*Real Estate Loans*	*Liquid Asset1*	*Liquid Asset2*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Crisis Exposure	0.028**	0.038**	0.023*	0.021*	0.001*	−0.018**	0.012**	0.017*
	(2.29)	(2.27)	(1.93)	(1.76)	(1.69)	(−2.15)	(2.26)	(1.80)
$Return_{1998}$		0.416*						
		(1.83)						
$Size_{2006}$		−0.059**	−0.029	−0.016	−0.000	−0.049***	−0.001	−0.007
		(−2.67)	(−1.57)	(−0.71)	(−0.13)	(−5.78)	(−0.19)	(−0.81)
$Return_{2006}$		−0.405**	−0.193	−0.216*	0.000	0.131*	−0.075	−0.136
		(−2.58)	(−1.55)	(−1.83)	(0.07)	(1.80)	(−1.51)	(−1.53)
BM_{2006}		−0.556***	−0.468***	−0.346***	−0.029***	−0.021	0.026	−0.039
		(−4.49)	(−4.24)	(−2.86)	(−4.16)	(−0.27)	(0.62)	(−0.52)
$Beta_{2006}$		0.114***	0.077**	0.094***	−0.001	−0.002	0.015*	0.031*
		(3.35)	(2.53)	(2.88)	(−0.55)	(−0.17)	(1.70)	(1.97)
$Book\ Leverage_{2006}$		−1.516	−1.999**	−1.113	0.062	−0.649	0.904***	1.322***
		(−1.38)	(−2.72)	(−1.59)	(0.85)	(−1.19)	(3.12)	(2.83)
$Deposit_{2006}$		−0.551*	−0.424	−0.197	0.001	−0.143*	−0.198	−0.486**
		(−1.95)	(−1.61)	(−0.76)	(0.07)	(−1.73)	(−1.64)	(−2.05)
$Tier\text{-}1\ Capital_{2006}$				1.184**	0.016	−1.123*	1.324***	1.798**
				(2.17)	(0.34)	(−1.87)	(3.21)	(2.73)
$Bad\ Loans_{2006}$				−15.853**	−0.568*	7.147***	−5.151***	−8.702***
				(−2.48)	(−1.87)	(3.75)	(−3.72)	(−3.90)
Education				0.002	−0.001	−0.007	−0.005	0.004
				(0.05)	(−0.37)	(−0.41)	(−0.42)	(0.18)
CEO Age				−0.000	−0.000	−0.000	−0.000	−0.000
				(−0.02)	(−1.13)	(−0.42)	(−0.59)	(−0.37)
Female				−0.048	0.003	−0.067	0.011	0.034
				(−0.85)	(0.46)	(−1.55)	(0.53)	(0.95)
Constant	−0.304***	2.631**	2.462***	1.143	−0.024	2.040***	−0.577	−0.501
	(−6.51)	(2.03)	(3.21)	(1.42)	(−0.29)	(3.04)	(−1.54)	(−0.80)
State FE	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Observations	252	181	252	252	252	229	229	229
R^2	0.011	0.184	0.363	0.400	0.432	0.514	0.506	0.484

This table shows results from cross-sectional regressions of stock returns for banks during the 2007–2008 financial crisis on their CEOs' S&L *Crisis Exposure*. We estimate
$Y_i = \alpha + \beta_1 Crisis\ Exposure_c + \lambda_1 X_{i,2006} + \lambda_2 C_c + f_s + \epsilon_i$.
The dependent variable in columns (1)–(4) is *FC Return*, which is the buy-and-hold return on the bank's stock from December 2006 through December 2008. All dependent variables in columns (5)–(8) denote their respective average value over the financial crisis years 2007 and 2008. In column (5), the dependent variable Y is *ROA*, which is the ratio of income before extraordinary items to assets. In column (6), Y is *Real Estate Loans*, which is the ratio of loans secured by real estate to assets. In column (7), Y is *Liquid Asset1*, which is the ratio of the sum of held-to-maturity securities and available-for-sale securities to assets. In column (8), Y is *Liquid Asset2*, where liquid assets include held-to-maturity securities, available-for-sale securities, cash and federal funds sold. CEO controls include education, age gender (Female). The bank control variables are measured as of the end of 2006; they include size (i.e., the natural log of the book value of assets), the annual stock return (*Return*), book-to-market ratio (*BM*), market beta, book leverage ratio, deposit ratio, tier-1 capital ratio and the fraction of non-performing loans (*Bad Loans*). The additional control variable in column (2) is the bank's stock return during the crisis of 1998. All variables are defined in Appendix A2. Columns (3)–(8) include state fixed effects. Standard errors are robust to heteroskedasticity and are clustered at the level of the bank's headquarter state. Numbers in parentheses are *t*-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

with credit risks and to be more aware than unseasoned CEOs of deteriorating loan quality. We measure credit risk with the share of loans that are past due by 90 days or more and nonaccrual loans. As these two columns show, *Crisis Exposure* is negatively correlated with loan quality and credit risk. With regard to economic magnitudes, a one–standard deviation increase in *Crisis Exposure* is associated with a 22 and 23 basis points decrease in *Bad Loans* and *Bad Loans/Loans* (respectively). Again, these findings are not driven by the unusual period of the 2007–2008 Financial Crisis as shown in Table A4 in the Appendix.

One caveat is that the observed lower credit risk associated with *conservative* CEOs can be due to more effective risk management or to less risk taking. It could be that *conservative* CEOs set up a more efficient and stronger risk control system and exert stringent monitoring of loan quality. Alternatively, managers may avoid risk and cut lending so as to minimize the possibility of being fired by the board; in this scenario, risk aversion reduces bank value. Yet we find in Section 5.1.1 (as a preview) that bank performance is positively associated with *Crisis Exposure* during crisis years and the two have an insignificant positive relation during non-crisis years. The implication is that the reduced risk taking associated with *conservative* CEOs does not reduce bank profitability or erode shareholder wealth. When we change the denominator from book value of total asset to total amount of loans, the negative relation between *Crisis Exposure* and *Bad Loans/Loans* still holds as seen in column (6). To the extent that *Bad Loans/Loans* can measure ex-ante lending quality, column (6) is another piece of evidence in favor of effective risk management by *conservative* CEOs.

In short, CEOs with a more severe experience of the S&L banking crisis are associated with lower credit risks and more effective credit risk management when they subsequently manage banks.

Overall, Section 4.1.2 illustrates that *conservative* CEOs learn from the specific causes of the S&L crisis and take precautions in the corresponding policies.

4.1.3. Endogeneity concerns

We now address endogeneity concerns. The ideal experiment for testing the causal effect of crisis experience on banks would involve the exogenous assignment of crisis shocks to identical CEOs randomly paired with identical banks. Such an assignment is impossible in reality. We must therefore deal with the impairment of any causal claims that are due to possible matching between the employment circumstances of individuals during the S&L crisis and their subsequent selection into banks. We leverage several approaches to isolate matching processes from the identification and thereby more firmly establish the effect of crisis experience.

CEO-bank matching after the S&L crisis. We first address a common limitation of CEO-related research—the matching between CEO and bank is not random. The concern is that the reported findings might result not from our hypothesized mechanisms but rather from unobservable factors that drive both the matching and our predicted variables.

To mitigate this concern, we first show that the findings do not reflect the continuation of a pre-existing trend for banks hiring *conservative* CEOs. We perform an event study of CEO turnovers which result in higher *Crisis Exposure* and examine the dynamic effect. We estimate a modified version of Eq. (2), where we include a set of dummy variables that identify years relative to the year a CEO with higher-than-predecessor *Crisis Exposure* takes office for an 8-year window surrounding the event. We set the base year as the year preceding the event by omitting the dummy for that year. The estimation uses a fully-saturated model and accounts for bank fixed effects, year fixed effects and bank- and CEO-level controls. Fig. 3 and Table A5 report the coefficient estimates. Across all panels with different bank policies as dependent variables, none of the coefficient estimates for the pre-event year dummies are statistically significant and we observe statistical significance only for post-event year dummies.

We then augment this event study by employing other CEO turnovers as a benchmark. The augmented test includes all turnovers and extends the list of covariates to include the interaction terms between year dummies (relative to the turnvoer year) and *Higher Exp.* which is a dummy variable for banks whose incoming CEOs have higher *Crisis Exposure* than outgoing CEOs. Table A6 reports the results. Each coefficient on the interaction terms indicates the difference in the outcome variable for banks with *Higher Exp.* equaling 1 between the given year and the year preceding CEO turnovers relative to the same difference for the benchmark banks. Across different dependent variables, the coefficient estimates on the interaction terms between *Higher Exp.* and pre-event year dummies are statistically insignificant and the statistical significance only holds for post-event interaction terms. These results further reject the possibility that pre-existing trends can explain our results and provide initial assurance that the CEO-bank matching is not driving the results.

Next, we exploit exogenous CEO turnovers to tackle the endogeneity concern. Since idiosyncratic death or health shocks to bank CEOs are too scarce to allow construction of a reliable regression sample, we rely on CEO retirement age to identify exogenous turnovers. This use of the retirement age is reasonable because a CEO's ideal retirement age is typically subject to debate, and the timing of CEO retirement is seldom exactly known in advance particularly when the CEO is elderly and more prone to sudden health issues. It follows that CEO replacement can be delayed or advanced despite the CEO succession plan. This variation in CEO selection due to semi-random timing allows us to causally identify CEO effects.

We classify the CEO turnover event as quasi-exogenous if the retired CEO is older than the sample median CEO retirement age of 63.[19] To ensure that older CEOs are not forced to exit because of bank policy changes, we verify the absence of significant trends in the years *before* the new CEO is appointed. We also find the abnormal cumulative return around these quasi-exogenous CEO turnovers to be indistinguishable from zero. We find that 43.5% of CEO turnovers in our sample are quasi-exogenous. This fraction is slightly lower than the literature (67.3% in Dittmar and Duchin, 2016), suggesting that age 63 is a conservative threshold.

Panel A of Table 4 reproduces the main findings with the subsample of exogenous CEO turnovers. Following Dittmar and Duchin (2016), we exclude the turnover year to avoid the potential disruptions accompanied by leadership changes. Panel A includes bank and state-year fixed effects as well as bank- and CEO-specific control variables.[20] The coefficients on *Crisis Exposure* estimated in the CEO exogenous turnover sample remain statistically significant. The interpretation is that banks reduce risk taking, interest rate risk and credit risk subsequent to appointing CEOs with a higher level of *Crisis Exposure*. These results offer further support for a causal interpretation of our results.

CEO staying with the same bank. If CEOs stay in the same bank from the time they witness a maximum state-year failure rate (that Crisis Exposure is based on) during the S&L crisis, their personal and banks' experiences of state-level shocks are not separable. To disentangle CEO crisis effects from bank effects, we restrict our attention to CEOs employed by other banks during the S&L crisis and thus remove 84 CEOs who stay with the same bank. We re-run the baseline panel regressions in Tables 2 and 3. Appendix Table A7 shows that CEO crisis effects continue to hold in this subsample.

A related concern is that CEOs' likelihood of staying with the same bank can be affected by banks' crisis experience which in turn explains future bank risk and performance, inducing estimation biases. However, we do not find empirical support for this possibility. First, we identify 161 banks that either entered CRSP or were founded before 1990 and construct their *Crisis Exposure* analogously. We do not detect any meaningful association between banks' *Crisis Exposure* and their likelihood of promoting S&L employees as CEOs using a Probit (OLS) model controlling for year fixed effects and bank and CEO characteristics. Second, we retrieve from BoardEx all individuals with employment histories from 1985 to 1990. We construct their *Crisis Exposure* and a dummy that identifies whether in the period from 1995 to 2009 they become a CEO of their then employer during the S&L crisis. These two variables are not statistically correlated in either the BoardEx sample or our baseline sample, estimated from cross-sectional Probit models controlling for individuals' birth year, gender, education, and position during the S&L crisis. These results address possible confounding effects arising from CEOs staying with the same bank.[21]

CEO-state matching before the S&L crisis. We then address the CEO-state matching issue stemming from CEOs' endogenous mobility during the S&L crisis. For instance, a risk-averse CEO in Texas could anticipate the banking crisis in that state and therefore decides to move to a "safer" state in which banks take less risk. These risk-averse CEOs will tend to adopt conservative business models and loan policies after stepping into the CEO position. In that event, *Crisis Exposure* captures only the innate risk aversion of CEOs and so our findings should be attributed mainly to CEOs' risk attitudes and not their early-career banking crisis experiences. However, this account would entail a *positive* correlation between *Crisis Exposure* and risk taking, which is the opposite of what Sections 4.1.1 and 4.1.2 show. It might also be that CEOs in the most heavily hit states are shunned by the labor market and switch to other states during or following the S&L crisis. Thus the CEOs from those states would then be employed only by lower-quality banks, which are more likely to assume risks. This account would again entail a *positive* correlation between *Crisis Exposure* and bank risk taking, which likewise contradicts our findings.[22] Another possibility is that

[19] We also use the retirement age of 65 as a robustness check. While the baseline results continue to hold, the statistical significance drops in light of the further reduced sample size.

[20] Results are not affected by excluding the CEO turnover year and or the three years around CEO turnovers.

[21] We thank an anonymous referee for suggesting this discussion.

[22] Nevertheless, we further alleviate this concern by repeating our analysis on a subsample of bank CEOs who remain in a single state throughout the 1985–1990 period. The results are similar to those estimated from the entire CEO sample.

risk-seeking individuals may decide to work in risk-averse states to counteract their personal traits, which may yield a *negative* correlation between the S&L crisis exposure and future bank risk.

In seeking further support for a causal link, we change the location used *Crisis Exposure* from the state where the CEO worked to the state where she was born. Since one's birthplace is (from her perspective) a random assignment, it follows that the S&L crisis severity in a CEO's birth state more closely approximates a random assignment to CEOs across banks than does its severity in her employment state. According to hometown "bias" studies (Yonker, 2017; Jiang et al., 2019; Lim and Nguyen, 2020), individuals remain connected to their states of birth via information and social networks and thus CEOs are likely exposed to crises in their birth states. We collect birthplaces of bank CEOs from Marquis Who's Who and Google search. We are able to obtain these data for one quarter of our sample CEOs, and this rate of successful search is slightly lower than that in Bernile et al. (2017). We then construct their $Crisis\ Exposure_{Birth}$ by incorporating the highest bank failure rate in each CEO's birth state over the 1985–1990 period.

We then reproduce key findings while using $Crisis\ Exposure_{Birth}$ and report the results in Panel B of Table 4. All columns include state-year fixed effects and bank- and CEO-specific control variables.[23] As shown in Panel B, the coefficient estimate of $Crisis\ Exposure_{Birth}$ is statistically significantly in all columns except for column (5). It follows that banks led by CEOs whose birth states are characterized by a higher intensity of the S&L crisis take on lower risk and implement conservative policies. This test helps confirm that our main results are not driven by matching between CEOs and states during the S&L crisis.

As an additional test, we exploit the variation in *Crisis Exposure* resulting from CEOs' non-overlapping time windows associated with the same state during the S&L crisis. This "within-old-state" variation is conditional on individuals' decisions about their employment states during the S&L crisis and thus less confounded by the CEO-state matching issue.[24] Specifically, we repeat the baseline analysis with additional fixed effects of CEOs' employment state during the S&L crisis. Table A8 in the Appendix shows that the point estimates of *Crisis Exposure* remain qualitatively similar to the baseline results. Unsurprisingly, the statistical significance drops slightly as we remove relevant variations in *Crisis Exposure*.[25]

In short, Section 4.1.3 attempts to address the major endogeneity concerns. The results indicate that the link between *Crisis Exposure* and bank outcomes is more than a statistical association.

Taken together, Section 4.1 demonstrates that CEOs' banking crisis experience carries forward to the future and has a measurable influence on banking outcomes. Consistent with the domain-specific learning, *conservative* CEOs implement conservative policies pertaining to the causes of S&L crisis.

4.2. Does CEOs' banking crisis experience matter during crisis years?

The *learning hypothesis* implies that banks with greater exposure to the S&L crisis navigated subsequent crises better. In addition, the crisis performance essentially captures the ex-post realized systemic risk (Acharya et al., 2017). Given the strong empirical pattern documented in Section 4.1.1 that CEOs' *Crisis Exposure* explains the ex-ante measure of systemic risk, we expect the banks' performance during the financial crisis years of 2007 and 2008 to vary with CEOs' *Crisis Exposure* as well.

There are three arguments supporting a positive effect of *Crisis Exposure* on banks' crisis performance. First, *conservative* CEOs could have identified risks and started taking corrective actions before the onset of the financial crisis, when it was easier to offload holdings of real estate loans. Second, *conservative* CEOs may take precautionary actions against possible crisis episodes such as the liquidity dry-up seen in the 1980s and thus strengthen the safety of banks. Third, investors may price banks led by veteran CEOs at a premium during crisis years, as they observe the tail-risks neglected during non-crisis periods to materialize and thus engage in flight-to-quality behavior in the spirit of Gennaioli et al. (2012).

4.2.1. Do banks managed by conservative CEOs perform better during crisis years?

Stock returns. We first investigate the association between *Crisis Exposure* and banks' stock return performance during crisis years. We require banks to exist and CEOs to be in office as of December 2016. We observe 273 such banks, 252 of which have observable crisis performance. On this sample, we estimate a cross-sectional regression for the recent financial crisis period as follows:

$$Y_i = \alpha + \beta_1 Crisis\ Exposure_c + \lambda_1 X_{i,2006} + f_s + \epsilon_i. \tag{3}$$

In this expression, the subscripts i and c denote the bank and its CEO. f_s is state fixed effects. The main independent variable of interest is the bank CEO's S&L *Crisis Exposure*. We also control for bank characteristics at the end of the calendar year 2006 and other CEO characteristics. The standard errors are robust to heteroskedasticity and clustered at the state level.

Table 5 presents results in support of the *learning hypothesis*. The dependent variable in columns (1)–(4) is *FC Return* (where "FC" denotes the financial crisis), which is the buy-and-hold return on the bank's stock from December 2006 through December 2008. The positive coefficients on *Crisis Exposure* indicate that bank stock returns during crisis years increase with respect to its CEO's S&L crisis experiences. The effect is economically and statistically significant. In column (1), the point estimate of the coefficient on *Crisis Exposure* is 0.028 in the cross-section of banks. Based on the coefficient in column (2) which controls for bank financial characteristics that the literature has found to be determinative of bank stock returns, a one-standard deviation higher S&L *Crisis Exposure* of the bank CEO is associated with a 4.2% (=0.038×1.094) higher return during the 2007–2008 financial crisis.[26] Relative to the sample mean for the buy-and-hold financial crisis return of −39.5%, this represents an increase of 10.5%. To put this in perspective, the bank crisis return of a CEO with 7% S&L state bank failure rate (the 75th *Crisis Exposure* percentile) is 3.1% higher than that of a CEO with 3.1% failure rate (25th percentile). Columns (3) and (4) include state fixed effects that account for local conditions and help alleviate the concern that the experience effect is driven by an assertive matching between CEOs' working states during the S&L crisis and their new states during the 2007–2008 financial crisis. In the presence of state matching, *Crisis Exposure* can coincide with new states' characteristics which affect the bank outcomes. The results are robust to this demanding control. In column (4), where we further control for tier-1 capital ratio, *Bad Loans* and CEO's education, age and gender, we find economically and statistically similar results.

As for the control variables of bank financial characteristics, most of them have the expected signs and their coefficients are qualitatively similar to those reported in Fahlenbrach et al. (2012) and Ellul and Yerramilli (2013). Banks that did better during the 2007–2008 financial crisis have lower book-to-market ratio, higher exposure to the market, more tier-1 capital and fewer bad loans at the end of 2006. There is suggestive evidence that banks with higher annual returns, size,

[23] Bank fixed effects are dropped in Panel B due to a small number of CEO turnovers involved.

[24] At the CEO level, regressing *Crisis Exposure* on fixed effects of CEOs' employment state during the S&L crisis yields R^2 of 0.644, suggesting that the across-old-state (within-old-state) fraction of the total variation in *Crisis Exposure* is 64.4% (35.6%).

[25] We thank an anonymous referee for suggesting this test.

[26] The standard deviation of *Crisis Exposure* is 1.094 during the 2008 financial crisis, as reported in Table 1 Panel B.

leverage and deposits at the end of 2006 underperform during the financial crisis.[27] In addition, the number of observations is lower in column (2) because they require the availability of 1998 bank stock returns as a control variable. In line with Fahlenbrach et al. (2012), a one-standard deviation increase of bank returns during the crisis of 1998 that followed Russia's default is associated with a 5.7% (=0.416 x 0.138) higher return during the recent financial crisis. However, our setting differs from theirs: the 1998 crisis resulted in severe damage to banks but did not trigger systemic bank closures. The nature of the shocks was different in the 1998 and S&L cases and hence so were their effects on CEOs and banks. While Fahlenbrach et al. (2012) highlight the institutional memory and persistent risk culture of banks, we stress the CEO's learning experiences and their risk management effects on banks.

For robustness, columns (1)–(4) in Table A9 repeat the same set of tests with *FC Abnormal Returns* as the dependent variable. *FC Abnormal Return* is the accumulative difference from December 2006 through December 2008 between the annual bank stock return and the expected annual return from a market model, where the latter is estimated by regressing daily returns on the bank's stock versus a constant and daily returns on the value-weighted CRSP index. The coefficients on *Crisis Exposure* are all significantly positive. To alleviate the concern that some banks were delisted amid the recent financial crisis and hence have a shorter holding period for the return calculation, we replace the dependent variables with the *Annualized* buy-and-hold and abnormal return during the crisis years in columns (5)–(8) in Table A9 of the Appendix. The estimates remain qualitatively and quantitatively comparable to the outcomes reported in the corresponding columns (1)–(4) of Table 5. To control for time-varying local conditions (i.e. demand), we also estimate panel regressions explaining the relationship between bank returns and CEO S&L *Crisis Exposure* with state-year fixed effects only for the crisis years 2007 and 2008. Table A10 of the Appendix report consistent findings. It is noteworthy that among the sample banks, 25 were observed to have changed CEOs during the 2007–2008 crisis. We also run a subsample test and include only those banks that did not experience CEO turnover during the crisis years; the results remain unchanged.

Overall, these results are broadly supportive of the hypothesis that banks led by CEOs with great exposure to the S&L crisis performed better in the stock market during the 2007–2008 financial crisis.

Exposure to the 2008 financial crisis. The analysis so far has focused on the outcome variables of stock returns. Here we examine the operation characteristics of banks. If the banking crisis experience helps CEOs identify and manage their banks' risks more effectively during crisis years, then we expect their banks to have profitable and safe business operations. We focus on three areas: operating profitability as measured by ROA, loan portfolio risks by real estate loans and liquidity risks by holdings of liquid assets.

To test the conjecture, we estimate regression specification in the form of Eq. (3). Columns (5)–(8) of Table 5 provide evidence concerning the association between banks' exposure to the 2008 financial crisis and our measure of S&L crisis experience. All columns include state fixed effects. In column (5), the dependent variable is the average ROA over the crisis years 2007 and 2008. The significantly positive coefficient on *Crisis Exposure* suggests that banks led by *conservative* CEOs outperform with respect to operational profitability. The economic magnitude is considerably large. A one-standard deviation increase of *Crisis Exposure* is associated with a 0.11% ($=0.001 \times 1.094$) increase. Relative to the sample mean of ROA during crisis years, this effect represents a 21.9% increase of ROA. This evidence also suggests that the previous findings on stock returns are underpinned by banks' differences in fundamentals and not solely driven by investors' motives of flight-to-quality during crisis years.

In column (6), we turn the focus to banks' real estate lending to shed light on their choice of loan composition in connection with CEOs' *Crisis Exposure*. The dependent variable is the average ratio of loans secured by real estate to book value of assets over the crisis years 2007 and 2008. We are interested in this exposure for two reasons. First, the literature has documented that exposure to real estate loans is negatively correlated to banks' crisis performance (Huizinga and Laeven, 2012; Ho et al., 2016). Second, both the 1980s S&L crisis and the 2007–2008 financial crisis were preceded and partially triggered by imprudent real estate lending. Although the former crisis mainly stemmed from commercial real estate bubbles and the latter from residential lending, CEOs who have lived through the former may be more judicious regarding the risk real estate loans pose and may even be able to unwind the corresponding positions in time. Correspondingly, the coefficient on *Crisis Exposure* is significantly negative, indicating that *conservative* bank CEOs are more cautious regarding real estate lending (whose risks exploded during the 2007–2008 crisis).

We then examine bank's liquidity risk. We have three motivations. First, the corporate literature claims that holding large amounts of cash may demonstrate a strong aversion to risk and a more conservative management style (Malmendier et al., 2011; Schoar and Zuo, 2017; Bernile et al., 2017; Dessaint and Matray, 2017; Dittmar and Duchin, 2016). Since our hypothesis is based on the notion that early-career experiences with a banking crisis raise one's awareness of the risks that can trigger failures, a CEO's preference regarding the amount of liquidity buffer is also likely to be affected. Second, in the absence of binding rules in 2007 and 2008, banks used their own discretion in managing liquidity risk (albeit in accordance with the general regulatory principles). Their discretionary holdings of liquid assets during crisis years are therefore more informative about banks' intrinsic attitudes toward liquidity risks. Third, the management of liquidity has become a core part of banks' strategic planning and balance sheet management post 2008.[28] Given the vital role that liquidity positions played in the 2008 crisis, a close check is warranted on the ex-ante factors underlying cross-sectional differences in handling liquidity buffers at the bank level.

With regard to the composition of the liquidity buffer, we follow the spirit of the LCR's definition that stipulates that the ratio's numerator (i.e., high-quality liquid assets) be easily and quickly convertible into cash. In line with that definition, the numerator of *Liquid Asset1* consists of held-to-maturity securities and available-for-sale securities; that of *Liquid Asset2* is the same as *Liquid Asset1* but also includes cash and federal funds sold. Both ratios are scaled by the book value of all assets.[29]

[27] Consistent with Fahlenbrach et al. (2012), we find that banks with higher exposure to the market in 2006 outperform during the 2008 crisis. However, the average beta over the tenure up until 2005 for CEOs used in the financial crisis test does not hold a statistically meaningful relationship with banks' crisis returns.

[28] The Bank for International Settlements introduced the liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR) in 2010 as part of Basel III. This regulation was initiated in response to calls for a sound liquidity environment following the unprecedented evaporation of liquidity observed during 2007–2008 (Brunnermeier, 2009). Beginning 1 January 2015, banks were required to maintain a minimum LCR ratio of 60% and to reach 100% by 1 January 2019 via annual increments of 10%. Banks were also required to comply with the NSFR disclosure requirements no later than 1 January 2018. The NSFR is a critical component of Basel III. Its goal is to reduce the risk of bank failures and the possibility of broader systemic stress in case disruptions to a bank's regular funding sources erode its liquidity position. Basel III's liquidity rules mark the first time that banks were required to meet global and quantitative minimum standards for liquidity.

[29] Cornett et al. (2011), Acharya and Mora (2015) and Irani and Meisenzahl (2017) adopt similar constructions. Note that our liquidity measure is based on market value, just as with the accounts reported in FR Y-9C statements. It would be ideal if we could separate the market valuation change of liquid

Table 6
CEOs' S&L crisis exposure and bank failures during the 2007–2008 financial crisis.

	Failure			
	(1)	(2)	(3)	(4)
Crisis Exposure	−0.023*	−0.021**	−0.045**	−0.046**
	(−1.92)	(−2.01)	(−2.15)	(−2.28)
$Size_{2006}$		0.017*	0.032*	0.041**
		(1.95)	(1.94)	(2.22)
$Return_{2006}$		−0.159*	−0.382*	−0.394*
		(−1.77)	(−1.80)	(−1.91)
BM_{2006}		0.169**	0.366**	0.411**
		(2.37)	(2.36)	(2.20)
$Beta_{2006}$		0.013	0.029	0.035
		(0.69)	(0.74)	(0.86)
$Book\ Leverage_{2006}$		0.753	1.373	1.970
		(1.10)	(1.05)	(1.37)
$Deposit_{2006}$		0.097	0.097	0.187
		(0.80)	(0.34)	(0.66)
$Tier\text{-}1\ Capital_{2006}$				0.692
				(0.79)
$Bad\ Loans_{2006}$				−5.824
				(−0.88)
Education				−0.013
				(−0.25)
CEO Age				0.003
				(0.94)
Female				0.000
				(.)
State FE	No	No	Yes	Yes
Observations	273	273	273	273

This table reports marginal effects from cross-sectional Probit regressions explaining the relationship between bank failure during the 2007–2008 financial crisis and CEOs' *Crisis Exposure*. The dependent variable is the *Failure* indicator (see text and Table A11 for details). CEO controls include education, age gender (Female) . The bank control variables are measured as of the end of 2006; they include size (i.e., the natural log of the book value of assets), the annual stock return (*Return*), book-to-market ratio (*BM*), market beta, book leverage ratio, deposit ratio, tier-1 capital ratio and the fraction of non-performing loans (*Bad Loans*). Columns (3) and (4) include state fixed effects. All variables are defined in Appendix A2. Standard errors are robust to heteroskedasticity. Numbers in parentheses are z-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

Table 7
Endogeneity Test for the Global Financial Crisis Results.

	(1) *FC Return*	(2) *ROA*	(3) *Real Estate Loans*
Crisis Exposure	0.044*	0.004*	−0.040**
	(1.88)	(1.75)	(−2.16)
Bank Controls	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Observations	112	112	101
R^2	0.631	0.550	0.557

This table tabulates the cross-sectional regressions results from the endogeneity tests (described in Section 4.2.3) for Table 5 and examines the subset of CEOs who were in office during the 2007–2008 Financial Crisis and were hired long before the crisis. The dependent variable in column (1) is *FC Return*. The dependent variables in columns (2) and (3) are the average *ROA* and *Real Estate Loans* over the crisis years 2007 and 2008. CEO controls include education, age gender (Female). The bank control variables are measured as of the end of 2006; they include size (i.e., the natural log of the book value of assets), the annual stock return (*Return*), book-to-market ratio (*BM*), market beta, book leverage ratio, deposit ratio, tier-1 capital ratio and the fraction of non-performing loans (*Bad Loans*). All variables are defined in Appendix A2. The regressions in Panel A include state fixed effects. Standard errors are robust to heteroskedasticity and are clustered at the level of the bank's headquarter state. Numbers in parentheses are *t*-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

Columns (7) and (8) of Table 5 present the regressions of liquid asset holdings. The results indicate a strong positive relations between liquidity positions and the crisis exposure to 1980s bank failures encountered by bank CEOs. Higher liquidity holdings would have served as a buffer amid the "black swan" event of the 2008 financial crisis. The economic magnitudes are comparable in size across the two columns. Since the mean of *Liquid Asset1* (*Liquid Asset2*) is 17.2% (31.8%) during the 2007–2008 financial crisis, it follows that a one–standard deviation increase in *Crisis Exposure* raises liquid asset holdings by 7.6% (5.8%).

Overall, *conservative* CEOs enhanced the asset safety of their banks' balance sheet during the subsequent 2008 crisis, especially when it came to items that have previously contributed to the cause and escalation of the 1980s S&L crisis. This evidence, consistent with domain-specific learning, indicates that bank CEOs have acquired industry- and crisis-specific insights from the past banking crisis, which they subsequently apply when facing the next calamity.

4.2.2. Are the banks led by conservative CEOs less likely to fail during crisis years?

We then address whether CEO banking crisis experience mitigates the likelihood of banks failing in subsequent crises. Table A11 in the Appendix tabulates the number of banks that survived and failed during the period from December 2006 through December 2009. We observe 273 sample banks as of December 2006. Following Fahlenbrach et al. (2012), we classify 18 banks as having failed during the financial crisis. Of these 18 failed banks, 7 were acquired by the FDIC. Four banks were merged at a discount, which we define as having happened if the price per share paid during the transaction is lower than the target's closing price on the last day prior to the merger announcement. Seven banks were forced into delisting by an exchange because they failed to meet the latter's requirement. We identify 255 banks as having survived the crisis. Among those, 230 banks remained listed on a US stock exchange in December 2009. Twenty-two banks were merged at a premium.

assets from the bank's propensity to hold liquid assets, but data constraints prevent this. Certain categories of liquid assets can appreciate or depreciate dramatically absent active bank management; as a result, a bank's liquid asset market value does not strictly reflect its willingness to hold a liquidity buffer and is confounded by its asset-picking skills. A liquid assets measure based on turnover or flow would be more appropriate but is precluded by data limitations. Fortunately, this concern is attenuated by the relatively stable pricing of liquid assets.

Three banks voluntarily delisted, citing the burden of compliance costs as the reason.

Empirically, we estimate a cross-sectional Probit model for the 2007–2008 financial crisis. The dependent variable *Failure* takes the value 1 if the bank failed between 2007 and 2009 (and takes the value 0 otherwise).[30] Table 6 reports the results and all columns report marginal effects. Columns (1) and (2) exclude state fixed effects and columns (3) and (4) include them (despite the potential "incidental parameters" problem explained by Greene). The empirical results indicate a negative relationship between the exposure of banking crisis experiences and the likelihood of subsequent bank failure. This negative correlation is consistently significant across different specifications and the economic magnitude is nontrivial. For the specification in column (2), at the mean level of *Crisis Exposure*, a marginal increase in crisis experience during the 1980s is associated with a 2.3% (=0.021 × 1.094) *lower* probability of failure during the 2007–2008 financial crisis. Given that the average probability of failure in our sample is 6.6%, this value corresponds to an economically significant decline of 34.8% in bank failure rates.[31] Comparing columns (3) and (4) with column (2), the effect size of *Crisis Exposure* more than doubles. Unsurprisingly, larger banks are more likely to fail, as are banks with fewer investment opportunities (captured by a higher book-to-market ratio).

Overall, Table 6 shows that banks led by CEOs with higher banking *Crisis Exposure* are *less* likely to have failed during the 2007–2008 financial crisis—conditional on a bank's past financial variables and some CEO characteristics. This finding corroborates the return results reported in Table 5 and supports the view that banks' crisis performance benefit from their CEOs' prior experiences in the S&L banking crisis.

4.2.3. Endogeneity tests

Since the cross-sectional tests during crisis years do not control for the bank heterogeneity, the results presented can be compatible with the alternative explanations as discussed in Section 4.1.3. This section attempts to mitigate this concern.

The first concern is that boards may select *conservative* CEOs to cope with the 2007–2008 financial crisis, rather than CEOs themselves affecting banks' crisis performance. To address it, we check those CEOs who are hired long before the onset of the 2007–2008 financial crisis and weathered its duration. Since banks cannot know ex-ante the arrival and the timing of the 2007–2008 financial crisis when hiring CEOs, these banks' boards are less likely to select CEOs' *Crisis Exposure* and appoint them in anticipation of this future crisis. In other words, the assignment of *Crisis Exposure* to banks during the recent crisis is plausibly orthogonal to boards' decisions for these cases. Empirically, we partition CEOs managing banks during the recent crisis into recently and distantly hired CEOs based on the median distance between 2007 and the year CEOs were appointed (i.e., 7 years). We classify 112 financial crisis CEOs who took office prior to 2000 as distantly hired CEOs. We repeat the analyses in Section 4.2.1 on this subset and summarize the results in Table 7. The enhancing effect of *Crisis Exposure* on crisis performance continues to hold, suggesting that CEO-bank matching does not solely drive our results.

Second, we allay the concern that CEOs staying with the same bank may induce estimation biases. Appendix Table A12 removes 55 "stayer" CEOs and repeats the financial crisis tests in Table 5. The results overall remain robust (although the statistical power in column (2) drops).

In light of the results so far (Section 4.2), we conclude that a higher prior banking crisis exposure of a CEO is associated with better crisis performance, spanning the stock market valuation, the soundness of business operations and the likelihood of failure during the crisis years.

[30] We also repeat the analysis using wider definitions of failure that includes either banks merging at a low premium or those requiring TARP funds at least twice during 2007–2009 and obtain similar results.

[31] Fahlenbrach et al. (2012) report a 7.5% failure rate during 2007–2009 in their sample of 321 banks.

Table 8
Heterogeneity Effects of CEO S&L Crisis Exposure.

	(1) *Beta*	(2) *Interest ROA Beta*	(3) *Bad Loans/Loans*
Crisis Exposure	−0.021	−0.063**	0.000
	(−0.74)	(−2.00)	(0.39)
*Crisis Exposure*S&L Bank Executive*	−0.098*	−0.194***	−0.003*
	(−1.96)	(−2.97)	(−1.72)
Bank Controls	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes
State×Year FE	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
Observations	2531	2350	2224
R^2	0.874	0.663	0.819

This table shows the interaction effect between CEOs' positions held during the S&L crisis and their S&L *Crisis Exposure* in explaining outcomes of their current employer banks. The table focuses on the 1995–2009 period. *S&L Bank Executive* is a dummy variable that identifies bank CEOs who already held a bank C-suite position during the S&L crisis. All regressions control for the same bank and CEO characteristic as in column (1) of Table 2, which are not reported to conserve space. All variables are defined in Appendix A2. The regressions include state-year and bank fixed effects. Standard errors are robust to heteroskedasticity and are clustered at the bank level. Numbers in parentheses are *t*-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

4.3. Salience of the S&L crisis experiences

Sector and Position Held during the S&L Crisis. The experienced-based learning of crises by an individual can vary as a function of salience. When the S&L crisis breaks out, people who work in the S&Ls, thrifts, banks and other depository institutions would be exposed to the shocks of institutional closures more than those working in other sectors. Bank failures are more salient for people within the banking sector because these people have larger networks, information sources and business connections related to it. Along the same lines, an individual who has held C-suite positions in a financial institution during the S&L crisis is likely to have faced precarious situations more often and vividly than if she were located near the bottom of the corporate ladder. Hence the learning effect on individuals is expected to be stronger in the former case than in the latter. Combining these two possibilities, we test whether the key results are stronger for those bank CEOs who were already C-suite executives in the banking sector when the S&L crisis occurred (1985–1990). To this end, we construct a dummy *S&L Bank Executive* which identifies that 47.2% of our sample CEOs were such; we then interact this dummy with *Crisis Exposure* in explaining banks' risk profiles.

Table 8 tabulates the results. It focuses on the 1995–2009 period and parallels columns (2) of Table 2 and columns (2) and (6) of Table 3. The negative significant coefficient on the interaction term *Crisis Exposure∗S&L Bank Executive* means that the dampening effect of *Crisis Exposure* on banks' risk taking, exposure to interest rate fluctuations and credit risk is amplified for the sample CEOs who worked as a bank executive when the S&L crisis occurred compared to the rest. Hence, Section 4.3 shows that *Crisis Exposure* matters even more when the bank failures are more salient for the sample CEOs in the S&L crisis.

5. Discussion and robustness

Section 5.1 discusses alternative explanations for the results in Section 4. Section 5.2 explores the sample selection issue. Section 5.3 lists a battery of robustness tests.

5.1. Alternative incentives

5.1.1. Destructive risk aversion

So far, our findings suggest that *conservative* CEOs have more conservative styles with respect to risk taking. The relevant welfare question is whether such risk avoidance creates or rather destroys shareholder

Table 9
Relationship between CEO S&L Crisis Exposure and Stock Returns.

	Return		*Abnormal Return*	
	(1)	(2)	(3)	(4)
Crisis Exposure	0.020		0.021	
	(1.18)		(1.37)	
*Crisis Exposure*FC*		0.044*		0.044*
		(1.90)		(1.66)
Crisis Exposure(1-FC)*		0.014		0.017
		(0.85)		(1.03)
Bank Controls	Yes	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes	Yes
State×Year FE	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Observations	2531	2531	2531	2531
R^2	0.662	0.662	0.654	0.654

This table reports panel regressions explaining the relationship between banks' stock returns and CEOs' S&L *Crisis Exposure*. The panel data include one observation for each bank–year pair and spans the 1995 to 2009 period. The dependent variable in columns (1)–(2) is *Return*, which is the buy-and-hold return on the bank's stock over the calendar year. In columns (3)–(4), the dependent variable is *Abnormal Return* which is the difference between *Return* and the expected return from a market model, where the market model is estimated by regressing daily returns on the bank's stock versus a constant and daily returns on the value-weighted CRSP index. *FC* is a dummy variable that identifies the financial crisis years 2007 and 2008. All regressions control for the same bank and CEO characteristic as in column (1) of Table 2, which are not reported to conserve space. All variables are defined in Appendix A2. All specifications include state-year and bank fixed effects. Standard errors are robust to heteroskedasticity and are clustered at the bank level. Numbers in parentheses are *t*-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

value. If CEOs refrain from assuming a level of risk exposure that is optimal for the banks they manage, then the risk-averse patterns documented here are destructive of value. For instance, a CEO is so concerned about her career that she takes no risks that could increase the likelihood of being fired; her strategy will not maximize shareholder value. If *conservative* CEOs do not reduce bank risk at the cost of forgoing investments with positive Net Present Value, then we expect that bank CEOs with higher *Crisis Exposure* to deliver equal or higher shareholder value as measured by the stock market performance.

We investigate the different value implications of *Crisis Exposure* during crisis versus non-crisis years. We define a dummy variable *FC* which identifies the crisis years 2007 and 2008, and we replace *Crisis Exposure* with two interaction terms—*Crisis Exposure∗FC* and *Crisis Exposure∗(1-FC)*—in explaining *Return* and *Abnormal Return*. Table 9 presents the results. All columns include state-year and bank fixed effects. In all columns, *Crisis Exposure∗FC* and *Crisis Exposure∗(1-FC)* are positively correlated with *Return* and *Abnormal Return*, suggesting that the average effect of *Crisis Exposure* is value-enhancing for banks.

We examine how the association between *Crisis Exposure* and stock returns differs between crisis and non-crisis years. Two comparisons are worth noting. First, the positive coefficient on *Crisis Exposure∗FC* is larger in magnitude and more significant statistically than the coefficient on *Crisis Exposure∗(1-FC)*. In fact, the positive coefficient on the latter lacks statistical significance. Second, the coefficient size of *Crisis Exposure∗FC* more than doubles that of *Crisis Exposure* in all columns. These two comparisons reveal that the value-enhancing effect of CEOs' S&L crisis experiences mainly manifests in subsequent crisis years rather than normal times and that these experiences do not hurt bank value during periods of normalcy. Regarding the potential explanation for why *Crisis Exposure* does not add as much shareholder value to banks in non-crisis years, *conservative* CEOs may shy away from the lucrative profits of derivative trading and imprudent real estate lending out of caution compared to their industry peers.

These findings suggest that the risk aversion tendencies we observe do not compromise shareholder value. Admittedly, the ideal test case would require constructing a counterfactual optimal risk level to be taken by banks—a project of doubtful feasibility.

5.1.2. Executive compensation

It is also possible that compensation packages provide CEOs with incentives that result in a range of preferences for risk (Fahlenbrach and Stulz, 2011; Graham et al., 2012; Frydman and Jenter, 2010; Ellul and Yerramilli, 2013). Therefore, it seems advisable to see whether the effect of banking crisis experience varies as a function of executive compensation. Compustat Execucomp has compensation data for 107 banks in our sample. For this subsample, we include the total compensation level as a control variable in our main specifications. We find that the sign of the *Crisis Exposure* coefficient is unaffected by controlling for CEO compensation. Owing to the decline in observations for the regression, statistical power lowers slightly when compensation controls are included. In addition, there exists no statistically meaningful association between *Crisis Exposure* and CEOs' salaries, bonuses, stock rewards, option rewards and the total compensation.

5.2. Sample selection

Our analyses so far focus on CEO cohorts who had employment history records during the S&L crisis and experienced it. To generalize the findings, we now extend the sample to include bank CEOs who did not experience the S&L crisis. We search from BoardEx for individuals who do not have employment history records during the S&L crisis and serve as a bank CEO during the 1995–2009 period. We identify 134 such bank CEOs. We assign the lowest value of *Crisis Exposure* to these additional CEOs and repeat the baseline analyses. Table A13 in the Appendix shows that the baseline results apply to this extended sample.

Exploiting this extended sample, we then perform matching-based tests to further mitigate the concern that the baseline findings are driven by CEO-bank matching as discussed in Section 4.1.3. Each CEO with S&L crisis experience (our baseline sample CEOs) has one no-experience control CEO matched on CEO age, education, bank size, return, beta and leverage in the year preceding the year when the former assumed office. The goal of this matching exercise is to ensure that the two groups share similar CEO characteristics and time-varying bank conditions which may drive banks' hiring decisions. Mahalanobis distance is used to identify the closest match and match with replacement. The matched sample includes 131 banks and the treatment and control groups have balanced characteristics prior to the turnover. We then compare post-turnover outcomes between the two groups by running panel regressions with state-year fixed effects on the matched sample. Table 10 tabulates the results. In Panel A, the independent variable of interest *CEO with Crisis Exposure* is a dummy variable that identifies CEOs with S&L crisis experience. In Panel B, *Crisis Exposure* takes the sample minimum value of *Crisis Exposure* for control CEOs. Across Panel A and B, banks managed by CEOs with (greater) exposure to the S&L crisis exhibit lower risk and more conservative policies. This evidence again suggests that CEO-bank matching alone does not drive our baseline results.[32]

A more challenging problem is that "potential CEOs" who lived through the S&L crisis may drop out of our sample. We first assess whether *Crisis Exposure* affects the likelihood of becoming a bank CEO. Specifically, we retrieve from BoardEx all individuals with employment histories from 1985 to 1990. We construct their *Crisis Exposure* and a dummy variable that identifies whether they become a bank CEO from 1995 to 2009. We do not detect a statistically meaningful association between the two using a regression framework. Another possibility is that the S&Ls crisis may slow down individuals' ascent toward CEOs and those who are more negatively affected will not have become a CEO by the test sample period of 1995 to 2009. However, we find that the age of becoming CEOs and *Crisis Exposure* have an insignificant correlation of 0.074. These evidence help alleviate the sample selection concern.

[32] We thank the editor and an anonymous referee for suggesting these tests.

Table 10
Matched Sample.

	Panel A: CEO with Crisis Experience					
	(1) *Beta*	(2) *MES*	(3) *Tail Risk*	(4) *Equity Volatility*	(5) *ROA Int. Beta*	(6) *Bad Loans/Loans*
CEO with Crisis Experience	−0.383***	0.007***	−0.005*	−0.050**	−0.080**	−0.002*
	(−5.47)	(3.52)	(−1.76)	(−2.39)	(−2.11)	(−1.70)
Bank Controls	Yes	Yes	Yes	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1349	1349	1349	1349	1228	1098
R^2	0.982	0.967	0.979	0.980	0.949	0.982
	Panel B: Crisis Exposure					
	(1) *Beta*	(2) *MES*	(3) *Tail Risk*	(4) *Equity Volatility*	(5) *ROA Int. Beta*	(6) *Bad Loans/Loans*
Crisis Exposure	−0.076***	0.001***	−0.001**	−0.010**	−0.021***	−0.001***
	(−4.52)	(3.22)	(−2.20)	(−2.55)	(−2.97)	(−3.51)
Bank Controls	Yes	Yes	Yes	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1349	1349	1349	1349	1228	1098
R^2	0.980	0.965	0.979	0.980	0.953	0.983

This table reports the key findings using a matched sample. Matching is performed on the extended sample including additional CEOs who did not experience the S&L crisis. Each CEO with S&L crisis experience (our baseline sample CEOs) has one no-experience control CEO matched on CEO age, education, bank size, return, beta and leverage in the year preceding the year when the former assumed office. In Panel A, the independent variable of interest CEO with Crisis Exposure is a dummy variable that identifies CEOs with S&L crisis experience. In Panel B, the key independent variable is *Crisis Exposure* which takes the sample minimum value of *Crisis Exposure* for no-experience control CEOs. All regressions control for the same bank and CEO characteristic as in column (1) of Table 2, which are not reported to conserve space. All variables are defined in Appendix A2. The regressions include state-year fixed effects. Numbers in parentheses are *t*-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

We also acknowledge that differential selection on unobservable characteristics could bias our estimation. However, the direction is ambiguous. For example, *Crisis Exposure* may capture local economic hardships which filter less capable individuals out of the bank CEO pool, inducing its positive correlation with unobservable talents that in turn explain our results. In this case, our baseline estimation represents an upper bound for the crisis experience effect. Alternatively, institutional failures during the 1980s may trigger individuals' concerns about the job insecurity of leadership roles and those who willingly become a bank CEO in the future are thus *less* risk-averse. This possibility implies that *Crisis Exposure* negatively correlates with CEOs' risk aversion and should increase future bank risk, thus biasing us from finding results.

5.3. Robustness checks

5.3.1. Other crises

Recall that the *Crisis Exposure* measure builds on the state-level bank failure rate CEOs witnessed during the S&L crisis. We now compare different types of crises affecting bank CEOs and their banks.

State versus bank level experiences. Recent papers show that the past employer-level distress and bankruptcy episodes affect the performance and risk attitudes of corporate managers and directors (Gopalan et al., 2021; Dittmar and Duchin, 2016). In case that the micro-level bank failures can subsume *Crisis Exposure* in affecting bank CEOs, we first place these two variables in a horse race and reproduce the major tests in Section 4.1. Toward that end, we create a dummy variable set equal to 1 when the institution employing CEO during the S&L crisis goes bankrupt (and set equal to 0 otherwise). In the full sample, roughly 4.7% of CEOs have experienced firm-level bankruptcy, similar to the median state-level failure rate. The odd columns of Table 11 tabulate the result. The result is that the economic and statistical power of the state-level *Crisis Exposure* survives the inclusion of employer shocks.[33]

[33] However, the variation in this firm bankruptcy indicator is probably insufficient for identification purposes, and the estimated coefficients are too noisy for drawing any reliable inference.

Table 11
Other types of CEO Crisis Exposure.

	Beta		*Interest ROA Beta*		*Bad Loans/Loans*	
	(1)	(2)	(3)	(4)	(5)	(6)
Crisis Exposure	−0.053***		−0.090**		−0.002*	
	(−2.65)		(−2.33)		(−1.75)	
S&L Bank Failure	0.047		−0.037		−0.011***	
	(0.46)		(−0.70)		(−2.66)	
$Exposure_{1997}$		−0.003		−0.013		0.001
		(−0.10)		(−0.93)		(1.19)
Bank Controls	Yes	Yes	Yes	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes	Yes	Yes	Yes
State×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2531	2345	2350	2199	2224	2051
R^2	0.864	0.863	0.624	0.613	0.788	0.779

This table shows results from regressions explaining the relationship between bank outcomes and different types of CEO crisis exposure. The table focuses on the 1995–2009 period. In odd columns, CEOs' S&Ls *Crisis Exposure* at the state level is compared with the bank level exposure during the S&L crisis in explaining the baseline results. *S&L Bank Failure* is a dummy variable that identifies the failure of banks CEOs worked for during the S&L crisis. In even columns, we conduct a placebo test using an exposure to a crisis unrelated to the US banking sector. *$Exposure_{1997}$*, analogous to *Crisis Exposure*, denotes the state-level bank failure rates during the 1997 Asian Financial Crisis. All regressions control for the same bank and CEO characteristics as in column (1) of Table 2, which are not reported to conserve space. All variables are defined in Appendix A2. The regressions in all columns include state-year and bank fixed effects. Standard errors are robust to heteroskedasticity and are clustered at the bank level. Numbers in parentheses are *t*-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

The 1997 Asian Crisis and 1994 Latin American Crisis. A natural question to ask is whether all economic recessions or asset bubbles can leave an imprint on future bank CEOs and/or influence their management styles or whether there was something unique about the S&L crisis. We address this question by constructing *Crisis $Exposure_{1997}$* which changes the "experience formation" period from the 1980s S&L crisis to the 1997 Asian crisis. We count bank failures in both 1997 and 1998

Table 12
Effects on Large Versus Small Banks.

	(1) *Beta*	(2) *Interest ROA Beta*	(3) *Bad Loans/Loans*
*Crisis Exposure*Small Bank*	−0.052**	−0.089**	−0.003**
	(−2.03)	(−2.28)	(−2.44)
*Crisis Exposure*Large Bank*	−0.058**	−0.083*	−0.002
	(−2.51)	(−1.96)	(−1.40)
Bank Controls	Yes	Yes	Yes
CEO Controls	Yes	Yes	Yes
State×Year FE	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
Observations	2531	2350	2224
R^2	0.866	0.625	0.787

This tables compares the effect of CEOs' S&L *Crisis Exposure* on large versus small banks. *Large Bank* (*Small Bank*) is a dummy variable that identifies banks with *Size* greater (not greater) than the sample median. The table focuses on the 1995–2009 period. All regressions control for the same bank and CEO characteristics as in column (1) of Table 2, which are not reported to conserve space. All variables are defined in Appendix A2. All regressions include state-year and bank fixed effects. Standard errors are robust to heteroskedasticity and are clustered at the bank level. Numbers in parentheses are *t*-statistics. We use ***, ** and * to denote statistical significance at the 1%, 5% and 10% level (respectively).

to allow sufficient time elapsed for the crisis consequences to emerge and also consider the 1998 LTCM crisis. Only four states have bank failure incidents (i.e., acquired by the FDIC or merged at a discount). In the even columns in Table 11, *Crisis Exposure*$_{1997}$ now captures the state-level bank failure rates where future bank CEOs were employed in 1997 and 1998. We find that exposure to this non–banking-related crisis does *not* affect CEOs in the same way as exposure to the S&L crisis did: the coefficient estimates reported in the even columns of this table are statistically insignificant for all the dependent variables. Similarly, we also re-build *Crisis Exposure* in connection with the 1994 Latin American crisis. Again, four states witnessed bank failures in 1994. We detect no predictive power of CEOs' exposure to the 1994 Latin American crisis either.

In sum, the S&L crisis is unique for bank CEOs due to its transformative and systemic impacts on the banking sector.

5.3.2. Large versus small banks

Gandhi and Lustig (2015) show that large and small banks are two different types in terms of the expected returns. To investigate whether CEO crisis experience exclusively affects large or small banks, we construct a dummy variable *Large Bank* (*Small Bank*) which identifies banks with size higher (lower) than the sample median and test the robustness of our main findings.[34] Table 12 reports the results. The coefficient estimate on the interaction term *Crisis Exposure∗Large Bank* is slightly larger (smaller) in the economic magnitude and statistical power than *Crisis Exposure∗Small Bank* in columns (1) ((2) and (3)). Therefore, we confirm that CEOs' S&L crisis experience works on both large and small banks, although the relative effect size is different between these two types for a given outcome variable.

5.3.3. Alternative measure construction and regression specification

Binary version of Crisis Exposure. To alleviate the concern of outliers driving the main results, we construct a binary variable *High Crisis Exposure* which identifies CEOs with *Crisis Exposure* higher than the sample mean. We re-estimate the main results in Section 4.1 using *High Crisis Exposure*. Table A14 shows that the predictive power of the binary version of *Crisis Exposure* continues to be strongly significant.

Alternative S&L period of Crisis Exposure. In case the choice of S&L crisis years drives the main result, we alter the definition of *Crisis Exposure* by expanding the S&L crisis period from the years 1985–1990 to 1980–1993. With the alternative definition, we repeat the baseline analyses and report the results in Appendix Table A15. None of our results are materially changed.

Alternative model specification. Table 6 exploits a Probit model to examine how *Crisis Exposure* affects bank failures during the 2007–2008 financial crisis. As a robustness check, Appendix Table A16 reports results estimated from OLS models with the most comprehensive set of control variables. Column (1) reports an OLS model without state fixed effect and column (2) with state fixed effects. Consistently, *Crisis Exposure* lowers the probability of bank failures during the crisis years.

Alternative sample period. Finally, one might question why we focus on the years 1995–2009 only. We also extend the sample period to 2015 and repeat the analysis in Section 4.1. Appendix Table A17 reports the results. The main conclusion continues to hold in the extended sample period.

[34] Alternatively, we define *Large* and *Small* relative to the within-year median of bank size and results remain similar.

6. Conclusion

This paper seeks to explain the substantial heterogeneity among bank risks and performance from the perspective of CEOs' early-career crisis experience. It also sheds light on how sophisticated agents learn from extreme events. We document that the S&L banking crisis experience of CEOs does affect their subsequent risk management style and crisis performance, and we offer suggestive evidence regarding the learning channel through which these effects work—namely the policies most likely to be affected pertain to the causes of the S&L crisis. We also leverage CEO birth town shocks and semi-exogenous CEO turnover to establish that our findings are not entirely driven by matching.

This paper is among the first to empirically confirm the role of CEO leadership in curtailing banks' systemic risk (ex-ante and ex-post) and that bank CEOs indeed learn from past banking crises. Identifying factors affecting bank risk in connection with CEO experiences can help shed light on building a financial system less fragile and vulnerable to macroeconomic shocks.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jfineco.2025.104009.

Data availability

Replication Package for "Do Bank CEOs Learn from Banking Crises?" (Reference data) (Mendeley Data).

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